Digital Ownership: The Tokenization of Real-World Assets

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Abstract

We study conditions under which tokenization creates value for indivisible real-world assets (RWAs). We show that tokenization does not create value merely by making an asset transferable, fractional, or digitally scarce. Value arises only when digital ownership records change the economics of ownership by leveraging on asset characteristics and past ownership records on the blockchain. This may generate retained value for past owners, create provenance value for later buyers, separate usage and financial rights, support membership benefits through fractional ownership, or strengthen post-sale incentives through royalties. These forces determine whether tokenized markets are inactive, thin but informative, lemons-like, or liquid but uninformative. We discuss implications for art and luxury tokens, private-equity and venture-capital tokens, real-estate tokens, tokenized claims on a social enterprise, and sustainable-firm tokens.

Keywords: Tokenization, blockchain, RWA, provenance, liquidity, royalties, private equity, real estate, art.

JEL Codes: D82, D86, G23, G32

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1 Introduction

The tokenization of real-world assets (RWAs) has become one of the most prominent proposed applications of blockchain technology in financial markets.[1] Tokenization—the creation of digital representations of ownership rights, cash-flow claims, usage rights, or other asset-linked claims on a shared ledger—is often associated with lower settlement costs, transparent ownership records, broader investor access, secondary-market tradability, fractional ownership, and programmable contractual features such as resale royalties. These features are commonly presented as improving liquidity and value, especially for assets that are difficult to trade in conventional markets. For public securities, Treasury instruments, or other already-traded claims, tokenization may primarily improve settlement, custody, or post-trade infrastructure. In contrast, this paper argues that for indivisible and thinly traded assets, tokenization may instead change the economics of ownership itself.

Examples include art and collectibles, real estate, private equity, venture-capital claims, entrepreneurial projects, privately held socially responsible firms, prestige and luxury assets, and other assets whose value combines financial payoffs with use, enjoyment, control, reputation, certification, affiliation, or status (Goetzmann et al., 2021; Hart and Zingales, 2017; Kitten and Chemla, 2026; Lovo and Spaenjers, 2018; Zingales et al., 2025). For these assets, ownership is not merely a claim to cash flows. It may also confer visibility, access, affiliation, prestige, or nonpecuniary utility from supporting a socially important or sustainable firm.

This paper develops a model that identifies when tokenization adds value for such indivisible, non publicly traded assets beyond basic operational efficiency gains.

Our model shows that the main channel of additional value creation for these indivisible and thinly traded assets is the degree of non-rivalry that digital records create, and we will give specific examples of the reasons for such non-rivalry below. We also show that in the absence of non-rivalry, i.e., under digital exclusivity, the assets may simply not trade,

[1] Recent market reports project substantial growth. Citi GPS (2023), for example, estimates that private-market tokenization could reach almost $4 trillion by 2030. On the economics and technological consideration of tokenization, see Canidio et al. (2022), Chemla et al. (2025), and Luo et al. (2026).

and features like royalties or fractional ownership may rather harm the value and liquidity than increase it. This observation may seem to be at odds with some common narratives, especially around non-fungible tokens (NFTs). However, it is also known that most NFTs never trade. Furthermore, one of the main platforms for creating and trading NFTs stopped imposing royalties (Kastrenakes, 2023). Our model helps to identify the conditions under which NFTs trade and royalties add value.

More generally, we study the main practical features of tokenization—traceability, tradability, fractionalization, separation of usage and financial rights, and royalties—and identify when they create value in equilibrium, and how these equilibrium results apply to the different asset classes considered.

The central economic feature of tokenization in our model is not digital scarcity per se, but the creation of a durable and verifiable ownership record. This record has two effects. First, it can change the information environment. When buyers observe past ownership and resale decisions, they may infer information about asset quality from the actions of earlier owners. This matters because informational asymmetries can depress prices, restrict trade, or eliminate gains from trade altogether, as in Akerlof's (1970) lemons problem, Milgrom and Stokey's (1982) no-trade theorem, and Myerson and Satterthwaite's (1983) impossibility result.

Second, the record can make ownership history payoff-relevant. If market participants care who previously owned an asset, the asset's provenance becomes part of its value. This is familiar in art markets, where ownership history affects authenticity, prestige, and willingness to pay, and where enjoyment value and resale considerations are central (Lovo and Spaenjers, 2018). Similar forces may arise for prestige or celebrity-owned real estate, early ownership of start-ups, claims associated with sustainable firms, or membership in visible investor communities. In particular, tokenized socially responsible investment can make the nonpecuniary component in Hart and Zingales (2017), Oehmke and Opp (2025), and Zingales et al. (2025) (more) persistent beyond the investor's holding period, because past ownership remains publicly observable.

We model an originator, Alice, who creates an indivisible asset and can issue a token. A first buyer, Bob, observes information about the asset's common value. Later buyers observe the ownership and resale record and may learn about the asset's common value from Bob's actions, with the informational content of these actions determined by equilibrium beliefs. The model separates tokenization's technical features from the economic mechanisms that make those features valuable. Traceability matters because it makes ownership and resale histories observable. Tradability matters only when resale creates gains from trade or reveals information. Fractionalization matters when a broader ownership base creates community, membership, or visibility benefits, but it may also reduce exclusivity. Royalties matter when they strengthen the originator's incentive to continue creating value after the initial sale; otherwise, they may simply tax secondary trade or be offset through the initial price. The framework therefore treats tokenization as a design problem: the originator may issue a bundled ownership token, separate usage and financial claims, fractional membership-like claims, or royalty-bearing claims. Which design is optimal depends on the source of value and the relevant friction.

Our baseline model considers two sources of non-rivalry. The first is *retained ownership* value: Bob may continue to benefit from having owned the asset even after selling it, for example because recorded ownership increases his reputation, status, or visibility. The second is *provenance* value: a later buyer may value the asset more because Bob previously owned it. These channels create gains from resale that would not exist in a purely rival ownership environment.

Conditional on non-rival value, the model generates distinct information regimes. When the asset is highly uncertain, Bob's purchase can be good news: the market is thin, but observed buying is informative. When uncertainty is intermediate, resale can instead be bad news because Bob sells after receiving unfavorable information. When uncertainty is low, or when retained-value and provenance effects are sufficiently strong, resale becomes liquid but uninformative because Bob always buys and resells regardless of his signal. Thus, tokenization changes not only whether an asset trades, but also what observed trades mean.

Liquidity and informativeness need not move together.

These regimes also yield asset-specific predictions. Venture-capital-like claims and many types of art are natural high-uncertainty cases in which early ownership may certify quality. Ordinary real estate, solar-energy firms, and social enterprises expanding access to healthcare and education often face a low degree of uncertainty, where resale may be relatively liquid but less informative. Distinctive real estate, celebrity-owned property, or culturally salient assets lie between these cases because their value depends partly on provenance. The framework therefore links the value of tokenization to the type of asset being tokenized, rather than treating tokenization as uniformly beneficial across RWAs.

We then use the framework to study alternative token designs. Separating usage and financial rights can create value by allocating different components of the asset to different buyers. This is natural for assets such as art and real estate, where display, occupation, enjoyment, and financial claims need not be held by the same person. Separation can also protect financial claims from negative provenance associated with prior use, although it may reduce value when positive provenance would otherwise support resale of a bundled token. Fractionalization creates another source of non-rival value. A wider ownership base may generate community, visibility, or membership benefits, but these benefits need not increase monotonically because excessive dispersion can reduce exclusivity. A well-known example is Bored Ape Yacht Club, where ownership of the digital token provides access to interaction with other members. More broadly, NFT markets suggest that many tokens never trade, while those associated with prominent early investor engagement may be more valuable (Oh et al., 2022). The model captures these effects by allowing ownership records and the size or composition of the ownership base to affect value.

Finally, we analyze royalties and post-sale incentives. Without post-sale effort, royalties are generally weakly dominated because they reduce resale incentives or can be offset by the initial price. Royalties become useful when the originator can take costly actions that increase the asset's future value, such as developing a project, maintaining a community, organizing events, preserving the asset, or increasing its visibility. In that case, royalties give

the originator a stake in secondary-market value and can improve incentives. The benefit is conditional: royalties are more attractive in informative-buy or liquid-resale equilibria, but may remain undesirable in lemons-market equilibria because they discourage trade.

The main message is that tokenization creates value only when its technical features map into economic mechanisms. Traceability can reduce adverse selection and create provenance. Tradability matters when transfers create non-rival value or reveal information. Fractionalization matters when shared ownership creates membership benefits without excessive dilution. Royalties matter when they strengthen post-sale incentives rather than merely taxing resale. The analysis therefore provides conditions under which tokenization improves liquidity, prices, and incentives for indivisible RWAs, and conditions under which it does not.

Related literature

Our paper contributes to several strands of literature. First, it relates to the growing theoretical literature on the economics of blockchain, smart contracts, and tokenization. Cong and He (2019) show how blockchain technology can expand the contracting space through decentralized consensus and smart contracts. Cong et al. (2021) study dynamic token adoption and valuation with network effects. Gryglewicz et al. (2021) analyze optimal financing with tokens when platforms face financing needs and moral hazard. Lee and Parlour (2022), Sockin and Xiong (2023) and Goldstein et al. (2024) analyze tokens as commitment devices in platform settings. Mayer (2021) studies token-based platforms in which transaction users and speculators interact. Li and Mann (2025) show how transparent token acquisition can support platform building by helping overcome coordination failures. Gan et al. (2021) study asset-backed ICOs and token design in the presence of moral hazard and demand uncertainty. Relative to this literature, we focus on tokenized indivisible RWAs rather than platform currencies, utility tokens, or network tokens. Our contribution is to show how digital ownership records affect learning from early informed investors' purchase decisions , adverse selection, provenance, resale, and post-sale incentives.

Second, the paper relates to work on adverse selection and information production

through markets and institutions. The lemons-market logic builds on Akerlof et al. (1970), while the possibility that observed actions reveal favorable information relates to models of herding, cascades, and rational learning (Banerjee, 1992; Cong and Xiao, 2024; Devenow and Welch, 1996; Tinn, 2010; Voorneveld and Weibull, 2011). The mechanism also connects to settings in which participation conveys information in crowdfunding and entrepreneurial finance (Åstebro et al., 2024; Brown and Davies, 2020; Chemla and Tinn, 2020; Jovanovic and Szentes, 2013). We contribute to the above by studying an environment in which the action is not only observed at a point in time but becomes part of the asset's persistent ownership history. With tokenization, who bought, who sold, and whether the asset was resold publicly verifiable are all potentially payoff-relevant. This can also help fund innovation in ways distinct from venture capital and other tradditional investment vehicles Casamatta (2003) and Kerr and Nanda (2015).

Third, our analysis connects to security design and corporate finance work on liquidity, control, and incentives. Gorton and Pennacchi (1990) show that composite securities can be valuable when some investors are privately informed. DeMarzo and Duffie (1999) develop a liquidity-based theory of security design, and DeMarzo (2005) studies pooling and tranching when intermediaries are informed. The idea that liquidity can weaken incentives or interact with control rights is central in Holmström (1979), Tirole (2010), and related work on skin in the game and moral hazard (Chemla and Hennessy, 2014, 2016). Our contribution is to show how digital ownership records and resale possibilities modify these trade-offs for indivisible RWAs. Tokenization changes both the information structure and the incentive value of secondary-market claims.

Finally, the paper is related to asset-specific work on art, real estate, collectibles, NFTs, and tokenized assets. Lovo and Spaenjers (2018) study trading in the art market, where ownership generates enjoyment value and resale considerations. Vorsatz (2019) analyzes collectibles tokenization and the separation of ownership and viewing rights. Empirical and descriptive work on NFTs and tokenized real estate highlights the importance of provenance, community, investor attention, and limited liquidity (Baum, 2021; Hardjono and Avrilionis,

2025; Nadini et al., 2021; Pinto-Gutiérrez et al., 2022; Smith and Baur, 2025). We contribute by providing a unified theoretical framework that clarifies when these asset-specific features create value and when they do not. In our setting, ownership history is not merely a record of past transactions; it can become part of the asset's future value through provenance, retained status, membership, and incentive effects.

The remainder of the paper is organized as follows. Section 2 introduces our stylized conceptual framework. Section 3.1 develops the baseline model. Section 4 analyzes token design. Section 5 discusses empirical predictions. Section 6 concludes. Appendix A constains the proofs.

2 Stylized conceptual framework

2.1 Environment and benchmark

We study the tokenization of real-world assets that are initially indivisible and lack a liquid secondary market in traditional environments. These include artwork, venture-capital or private-equity-backed business ideas or real estate. In what follows, we will refer to any of these as "the asset".

The asset is created by originator Alice. Here our focus is on ex ante creation incentives, rather than full surplus. The asset is assumed to be socially valuable and we take Alice's payoff as the welfare-relevant objective, reflecting the role of ex ante incentives to invest in its creation. After creation, Alice seeks to monetize the asset, and we normalize her utility from retaining the asset to zero.

In the no tokenization benchmark, Alice has an opportunity to sell the asset to a single buyer, Bob, at price p_A. Under a buyer's ownership, the asset generates value along two dimensions: an enjoyment (or use) value and a continuation value. In the benchmark, Bob is the relevant buyer, so we denote these by z_B and V, respectively. The continuation value V captures the future financial value associated with ownership, including the benefits of holding the asset over time, such as dividends or rental income when relevant, and the

expected payoff from a later exit, such as a sale or an initial public offering.

The benchmark assumes that there is no short-term secondary-market resale opportunity. The asset is indivisible over this horizon: if Bob buys it, he obtains both the ownership value and the use value and he must retain the asset for some time.

2.2 Tokenization and potential economic channels

Tokenization has the potential to modify this environment along several dimensions.

First, it may facilitate short-term secondary markets. We model this as the possibility that, after acquiring the asset, Bob may transfer some or all associated claims to another agent, Carol. We denote the price Bob charges Carol by p_B.

Second, tokenization may affect valuations through improved record-keeping and traceability of ownership, for example, via digital provenance.

Third, it allows ownership and use to be separated. The enjoyment and continuation components of value need not be jointly held and may be traded as separate tokens. This possibility may be particularly relevant for art and real estate assets, where the usage value can be substantial compared to the financial value. In that case, Alice may set separate prices for the ownership and usage tokens.

Fourth, tokenization expands the set of feasible allocations of claims. In particular, ownership need not be all-or-nothing and can be designed to allow fractional ownership, so that multiple agents hold claims to the same asset.

Fifth, contractual features such as royalties may be embedded at issuance via smart-contract functionality, enabling Alice to receive a share of prices in the short-term secondary market. This can further affect Alice's incentives to engage with value creation after the initial sale.

The main model developed in Sections 3 and 4 will consider these channels in sequence and introduce a specific information structure. Before doing so, it is useful to highlight some key considerations in reduced form. Let $\tilde{z}_B$ and $\tilde{z}_C$ denote post-trade enjoyment values, let V denote the benchmark continuation value under sole ownership, and let V_B and V_C denote

the corresponding continuation payoffs after trade between Bob and Carol. The allocation $(\tilde{z}_B, \tilde{z}_C, V_B, V_C)$ nests full transfer, partial transfer, and fractional ownership.

2.3 Preliminary implications

Suppose first that Alice creates an asset and sells it to Bob. If Bob retains the asset, his payoff is

$$z_B + V.$$

Now consider a secondary market in which Alice creates either one bundled token or separate tokens. In a frictionless market, secondary trade of a bundled token is efficient if

$$\tilde{z}_B + \tilde{z}_C + V_B + V_C \geq z_B + V,$$

If usage and ownership rights are tokenized separately, the corresponding efficiency conditions are $\tilde{z}_B + \tilde{z}_C \geq z_B$ for the usage token and $V_B + V_C \geq V$ for the ownership token.

These conditions imply that secondary markets may create value through two channels: a more efficient allocation of usage or enjoyment value, and a change in total continuation value. In particular, if the continuation value is rival and fixed, so that $V_B + V_C = V$, gains from trade arise solely from differences in usage or enjoyment value across owners. By contrast, a secondary trade creates additional value only if tokenization increases the total continuation value or the total usage or enjoyment value, so that $V_B + V_C > V$ and/or $\tilde{z}_B + \tilde{z}_C > z_B$. In other words, value creation requires some form of non-rivalry in usage value, continuation value, or both. We study possible sources of such non-rivalry, with particular emphasis on the record-keeping benefits of digital ownership. We also consider conditions under which tokenization creates value without full transfer, either through separate token rights or through fractional ownership and membership benefits.

Moreover, suppose Alice can set a royalty $r \in [0, 1)$ on secondary sales. Alice's expected

payoff from selling to Bob, followed by a possible resale from Bob to Carol, is

$$\mathbb{E}\left[p_A \cdot 1_{AB}\right] + r\,\mathbb{E}\left[p_B \cdot 1_{AB} \cdot 1_{BC}\right],$$

where 1_{AB} denotes the event that Bob buys from Alice, and 1_{BC} denotes the event that Bob successfully resells to Carol. Bob resells only if there exists a price p_B such that, at the secondary-trading date,

$$(1-r)p_B + \tilde{z}_B + V_B \geq z_B + V$$

for Bob, while Carol is willing to buy only if

$$\tilde{z}_C + V_C \geq p_B.$$

Analogous conditions apply when separate tokens are traded.

Absent gains from trade or value creation under resale, a positive royalty cannot be imposed without eliminating a secondary trade. To see this, consider the case in which the enjoyment value is certain and constant, and both enjoyment and the continuation value are rival: $\tilde{z}_B = \tilde{z}_C = z_B = \bar{z}$, $V_B = 0$, $V = V_C = \bar{V}$. The highest price Bob can charge Carol is then $p_B = \bar{z} + \bar{V}$. However, Bob does not resell at this price, because $(1-r)(\bar{z}+\bar{V}) < \bar{z}+\bar{V}$ for any $r > 0$.

Even when a resale occurs with a positive royalty, the royalty need not create additional value for Alice. If the resale price p_B is invariant to r, Alice can offset royalty revenue through the initial price p_A. To see this, suppose both purchase and resale occur with probability one, so that $1_{AB} = 1$ and $1_{BC} = 1$. Alice is then indifferent between setting $p_A = p_A^r$ with royalty $r > 0$, which yields $p_A^r + r\,\mathbb{E}[p_B]$, and setting no royalty but charging the higher initial price $p_A = p_A^r + r\,\mathbb{E}[p_B]$. The latter generates the same expected revenue without distorting secondary-market trade.

These observations imply that tokenization creates value beyond basic operational efficiencies only when its technical features map into economic mechanisms: gains from trade,

non-rival ownership benefits, information revelation, or incentive effects. For this reason, the baseline model abstracts from royalties. We return to royalties in Section 4.3, where Alice can affect asset value after the initial sale.

3 Baseline model

This section develops a tractable framework for studying the value drivers of tokenization identified above. We begin with a baseline setting in which financial and usage rights are bundled. In this baseline, tokenization creates value only through secondary-market trade in the bundled asset rights, represented by the possibility of resale from Bob to Carol (Sections 3.1–3.3). The analysis focuses on digital ownership records, Bob's private information about asset value, and how this information shapes secondary-market liquidity.

Our goal is to keep the setting flexible enough to accommodate different types of indivisible assets. These assets vary in their degree of uncertainty about common value and in the relative importance of usage or enjoyment value. For example, art and venture-capital-backed business ideas are likely to involve greater uncertainty than real estate projects. By contrast, usage and enjoyment value may be more important for art and real estate than for tokenized venture capital or private equity.

Section 4 then modifies the baseline assumptions to study the effects of separate token rights, membership clubs and fractional ownership, and royalties.

Throughout the analysis, we focus on pure-strategy Perfect Bayesian Equilibria (PBE) and assume, for clarity, that all agents are risk-neutral.

3.1 The setting

The asset. Alice creates an asset and her value from retaining it is normalized to zero. When owned by other agents, the asset generates a common financial value v. The asset is initially sold to Bob, who can be interpreted as a sophisticated buyer such as a venture capitalist, art curator, or especially enthusiastic collector. The common financial value is

not perfectly known in the short term and is drawn from a Beta distribution:

$$v \sim \text{Beta}\left(\lambda\bar{v}, \lambda(1-\bar{v})\right),$$

where $\bar{v} = \mathbb{E}[v]$ is the unconditional expected value. The parameter λ measures the precision of prior information: lower values of λ correspond to greater uncertainty, while $\lambda \to \infty$ corresponds to the limiting case of no uncertainty. We use the Beta distribution because it provides a tractable way to capture differences in uncertainty across indivisible assets while preserving analytical solutions.[2] For example, art and innovative business ideas are likely to involve lower values of λ than real estate or growth capital.

Bob receives a signal $s_B \in \{0, 1\}$ about the asset value, where $Pr(s_B = 1|v) = v$. Given the properties of the Beta distribution, Bob's posterior expectation of the common value is

$$\mathbb{E}[v|s_B] = \frac{\lambda\bar{v} + s_B}{\lambda + 1},$$

and the unconditional probability that Bob receives a high signal is $Pr(s_B = 1) = \bar{v}$. It follows that the magnitude of Bob's updating is $|\mathbb{E}[v|s_B] - \mathbb{E}[v]| = \frac{|s_B - \bar{v}|}{\lambda+1}$. Thus, Bob learns more from his signal when asset-specific uncertainty is higher, that is, when λ is lower. This signal is likely to be more important when information is soft and time-varying, as is likely to be the case for private equity and venture capital or sustainable firms with intangible capital, than when it is hard and valid across time, as in core real estate, art, and luxury goods, where information can be recorded and made publicly available on a blockchain. Further, the high degree of uncertainty in venture capital, highly-leveraged private equity investment, or in some types of art may exacerbate Bob's learning from his signal.

In addition to the common financial value, Bob and Carol may receive usage or enjoyment benefits from the asset. Let z_i denote the usage value of agent $i \in \{B, C\}$. This value is obtained only by an agent who uses or enjoys owning the asset; it is agent-specific and non-

[2]The Beta distribution has several convenient properties for our setting. It keeps asset value positive and bounded, accommodates skewed, U-shaped, and hump-shaped distributions, and is conjugate to Bernoulli signals. This helps preserve tractability in posterior beliefs and sale probabilities.

transferable. To keep the baseline setting simple, assume that the core usage value is known and that $z_B = \bar{z}$. If Carol buys the asset directly from Alice, then $z_C = \bar{z}$. If instead Carol buys the asset from Bob, her usage value may be affected by digital ownership records, as described below. Finally, we assume that the core usage value does not exceed the prior expected financial value: $\bar{z} \leq \bar{v}$.

Digital records. To capture the role of digital records as a potential source of non-rivalry, suppose that recording the tokenized asset on a blockchain or other digital ledger makes ownership and usage histories verifiable. This record can create additional benefits or costs. The baseline model considers two such effects.

The first is *retained value* for Bob after a secondary sale. Specifically, suppose that if Bob sells the asset to Carol in the secondary market, he retains value $\varsigma_B \geq 0$. This retained value reflects benefits from verifiable past ownership. For example, if ownership of digital art generates status, reputation, or public association with the artwork, then some of this value may persist after resale.

The same mechanism may apply to other asset classes. In venture capital and private equity, having identified and invested in a successful project may increase a fund's visibility and attract future deal flow. Established venture capitalists may already obtain these benefits through conventional reputational channels, but digital ownership records can make the association explicit and verifiable, especially for less established investors. By contrast, retained value is likely to be less important for ordinary tokenized real estate. For tractability, the baseline model treats ς_B as a known constant.

The second effect is *provenance*: past ownership or usage may affect the enjoyment value of later buyers in the secondary market. We model this effect as an exogenous constant δ_B that shifts Carol's usage value. If Carol buys the asset from Bob rather than directly from Alice, her usage value is $z_C = \bar{z} + \delta_B$. When $\delta_B \geq 0$, Bob's prior ownership increases Carol's enjoyment value. This may occur when Bob's identity, reputation, or public association with the asset is valued by later buyers. Art markets provide a natural example, since provenance is a standard component of valuation. Related effects may also arise in real estate, for

example when a property was previously owned or occupied by a celebrity. By contrast, $\delta_B < 0$ captures cases in which prior ownership or use reduces later enjoyment value. This is plausible for ordinary real estate, where used properties may be valued less than otherwise comparable new properties.

We view this exogenous provenance channel as less central for most private equity and venture-capital-like assets, where δ_B is likely to be close to zero unless investors enjoy funding a venture or firms have social value. This does not rule out the possibility that a venture capitalist's purchase increases the asset's perceived value. Rather, in our model such effects operate through the information channel associated with the common value v. Thus, any value-increasing effect of a venture capitalist's purchase is treated as an endogenous consequence of information revelation, rather than as an exogenous provenance or celebrity effect.

These assumptions should be interpreted as capturing the *relative importance* of common financial value and idiosyncratic usage value across asset classes. The model's results are invariant to proportional changes in scale, such as differences in the overall value of a particular artwork, real estate asset, or business idea.

Baseline game. We can now specify the baseline game and its timeline. Let $1_{AB} \in \{0, 1\}$ denote the event that Bob buys the asset directly from Alice, and let $1_{AC} \in \{0, 1\}$ denote the event that Carol buys the asset directly from Alice. Let $1^S_{AB} \in \{0, 1\}$ denote the event that Bob puts the asset up for resale, and let $1_{BC} \in \{0, 1\}$ denote the event that Carol buys the asset from Bob, conditional on it being available.

Alice chooses the primary-market price p_A. Bob decides whether to stay out, buy and keep the asset, or buy and offer it for resale at price p_B. Carol decides whether to stay out or buy the asset, either from Alice or from Bob.

Because Bob is privately informed about the common value v, Carol's beliefs about Bob's earlier actions affect her willingness to buy. Carol can buy from Bob only if Bob has bought the asset and offered it for resale. She can buy directly from Alice only if Bob has not bought the asset. We denote Carol's belief about the asset's common value by $\mu_{AC} = \mathbb{E}\left[v \mid \tilde{1}_{AB} = 0\right]$

when she can buy directly from Alice, and by $\mu_{BC} = \mathbb{E}\left[v \mid \tilde{1}_{AB} = 1, \tilde{1}^S_{AB} = 1\right]$ when she can buy from Bob. The tildes denote conjectured equilibrium actions used to form beliefs. In any PBE, these conjectures must coincide with equilibrium strategies on path.

The timeline is as follows:

- *Date 0.* Alice creates an asset with fixed fundamental characteristics. She sets p_A to maximize her expected payoff,

$$u_A = p_A \mathbb{E}\left[1_{AB} + (1 - 1_{AB})\, 1_{AC}\right] \tag{1}$$

  taking as given the parameters of the model and equilibrium beliefs about Bob's and Carol's strategies.

- *Date 1.* Bob observes his signal s_B and the price p_A. He decides whether to stay out, which gives utility zero, or to buy the asset. If he buys the asset, he subsequently decides whether to keep it or offer it for resale. If Bob buys and keeps the asset, his utility is

$$u^K_B(s_B) = \mathbb{E}\left[v \mid s_B\right] + \bar{z} - p_A. \tag{2}$$

  If Bob buys and offers the asset for resale at price p_B, his expected utility is

$$u^S_B(s_B) = \mathbb{E}\left[1_{BC}\left(p_B + \varsigma_B - p_A\right) + (1 - 1_{BC})\left(\mathbb{E}\left[v \mid s_B\right] + \bar{z} - p_A\right)\right]. \tag{3}$$

  Bob's optimal decision depends on Carol's equilibrium beliefs and purchase behavior.

- *Date 2.* Carol can buy the asset only if it is available, either from Alice at price p_A or from Bob at price p_B. Her utility is

$$u_C = \begin{cases} \mu_{AC} + \bar{z} - p_A, & \text{if she buys from Alice,} \\ \mu_{BC} + \delta_B + \bar{z} - p_B, & \text{if she buys from Bob.} \end{cases} \tag{4}$$

3.2 Benchmark

Before solving the full model, we first analyze the benchmark without tokenization. In this benchmark, Alice can sell the asset only to Bob. Bob observes the informative signal described above, but cannot resell the asset in the short term. Hence, there is no retained value and no provenance effect.

Moreover, ownership and usage rights cannot be traded separately. Bob therefore buys the asset if and only if $\mathbb{E}[v \mid s_B] + \bar{z} \geq p_A$. Alice chooses the price p_A to maximize $u_A = p_A \pi_B$, where π_B is the probability that Bob buys the asset.

Lemma 3.1. *In the benchmark without tokenization, Alice can sell the asset only to Bob in the short term. In the corresponding PBE, there exists a threshold*

$$\bar{\lambda}^{bm} \equiv \frac{\bar{v} - (1-\bar{v})\bar{z}}{(1-\bar{v})(\bar{z}+\bar{v})}$$

such that Alice sets $p_A = p_A^{bm}(0) = \frac{\lambda\bar{v}}{\lambda+1} + \bar{z}$ *if* $\lambda \geq \bar{\lambda}^{bm}$*, and* $p_A = p_A^{bm}(1) = \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}$ *if* $\lambda \leq \bar{\lambda}^{bm}$*. Under* $p_A = p_A^{bm}(0)$*, Bob always buys, so* $\pi_B = 1$*. Under* $p_A = p_A^{bm}(1)$*, Bob buys the asset with probability* $\pi_B = \bar{v}$*. Alice's equilibrium payoff is*

$$u_A^{bm} = \begin{cases} \frac{\lambda\bar{v}}{\lambda+1} + \bar{z}, & \text{if } \lambda \geq \bar{\lambda}^{bm}, \\ \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}\right), & \text{if } \lambda \leq \bar{\lambda}^{bm}. \end{cases} \tag{5}$$

Proof. Bob is willing to buy if and only if $\frac{\lambda\bar{v}+s_B}{\lambda+1} + \bar{z} \geq p_A$. If Alice sets $p_A > \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}$, Bob never buys and Alice's payoff is zero. If she sets $\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} < p_A \leq \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}$, Bob buys only after a high signal, $s_B = 1$. In this region, Alice optimally sets the highest feasible price $p_A^{bm}(1) = \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}$, and Bob buys with probability $\bar{v}$. If Alice sets $p_A \leq \frac{\lambda\bar{v}}{\lambda+1} + \bar{z}$, Bob buys after both signals. In this region, Alice optimally sets the highest feasible price, $p_A^{bm}(0) = \frac{\lambda\bar{v}}{\lambda+1} + \bar{z}$, and Bob buys with probability one. The threshold $\bar{\lambda}^{bm}$ is obtained by comparing Alice's payoff under the low-price strategy, $p_A^{bm}(0)$, with her payoff under the high-price strategy, $\bar{v}p_A^{bm}(1)$. Solving $p_A^{bm}(0) = \bar{v}p_A^{bm}(1)$ for λ gives the stated threshold. □

Before proceeding to the full game, it is useful to note the benchmark comparative statics. Alice's payoff is highest for assets with either low uncertainty, corresponding to high λ, or high uncertainty, corresponding to low λ. This follows (5): the low-price payoff is increasing in λ, while the high-price payoff is decreasing in λ.

For low-uncertainty assets, Alice optimally sets a price low enough that Bob buys regardless of his signal. As uncertainty increases, asymmetric information forces Alice to reduce this price. For high-uncertainty assets, by contrast, Alice optimally sets a higher price and sells only when Bob receives a favorable signal.

The benchmark payoff is therefore lowest at an intermediate level of uncertainty, around $\bar{\lambda}^{bm}$. Higher usage value $\bar{z}$ reduces the range of parameters for which the high-price strategy is optimal. Intuitively, usage value makes Bob's willingness to pay less sensitive to his signal, making it more attractive for Alice to sell to both signal types. These observations carry over to the tokenized environment, where information revelation and secondary-market liquidity also depend on the degree of asset-specific uncertainty.

3.3 Results: secondary market and liquidity

We now turn to the case in which tokenization allows Bob to resell the asset to Carol. We begin with a one-token environment without royalties. Carol does not receive a private signal, but her beliefs depend on Bob's equilibrium actions, which may reveal information about Bob's signal. The full characterization is provided in Appendix A.1; Proposition 3.2 summarizes the main equilibrium regimes.

Proposition 3.2. *Suppose that there is one bundled token that can be resold on the secondary market. The PBE that give the highest utility to Alice in each parameter region are as follows. (1) If $\delta_B + \varsigma_B \leq 0$, Bob always keeps the asset and the secondary market is inactive. This outcome coincides with the benchmark without tokenization.*

(2) If $0 \leq \delta_B + \varsigma_B \leq \frac{1-\bar{v}}{\lambda+1}$, there is a partially liquid market. Define

$$\bar{\lambda} \equiv \frac{\bar{v} - (1-\bar{v})(\bar{z} + \delta_B + \varsigma_B)}{(1-\bar{v})(\bar{z} + \bar{v} + \delta_B + \varsigma_B)}.$$

(2a) If $\lambda \leq \bar{\lambda}$, the equilibrium features informative purchase by Bob followed by resale, i.e., "informative buy". Bob buys the asset if and only if $s_B = 1$ and then resells it to Carol. Carol does not buy directly from Alice. The prices are $p_A = \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \varsigma_B$, and $p_B = \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B$, and Alice's expected payoff is

$$u_A = u_A^{IB} \equiv \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \varsigma_B\right). \tag{6}$$

(2b) If $\lambda \geq \bar{\lambda}$ then the equilibrium involves a lemons market. Bob always buys the asset, but resells it only after observing $s_B = 0$. The prices are $p_A = \frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \varsigma_B$, and $p_B = \frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B$, and Alice's expected payoff is

$$u_A = u_A^{Lem} \equiv \frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \varsigma_B. \tag{7}$$

(3) If $\delta_B + \varsigma_B \geq \frac{1-\bar{v}}{\lambda+1}$ then the market is liquid but uninformative: Bob always buys and always resells the asset. The prices are $p_A = \bar{v} + \bar{z} + \delta_B + \varsigma_B$, and $p_B = \bar{v} + \bar{z} + \delta_B$, and Alice's expected utility is

$$u_A = u_A^U \equiv \bar{v} + \bar{z} + \delta_B + \varsigma_B. \tag{8}$$

Proof. See Appendix A.1. □

Proposition 3.2 shows that there is value in creating a short-term secondary market only if there is non-rivalry, i.e., when $\delta_B + \varsigma_B > 0$. In contrast, when $\delta_B + \varsigma_B \leq 0$, the secondary market is inactive even though resale is technically feasible. This is because a resale has no surplus to allocate.

Provided that the combined effect of retained value and provenance is positive, the resulting market structure depends on the degree of asset-specific uncertainty. For highly

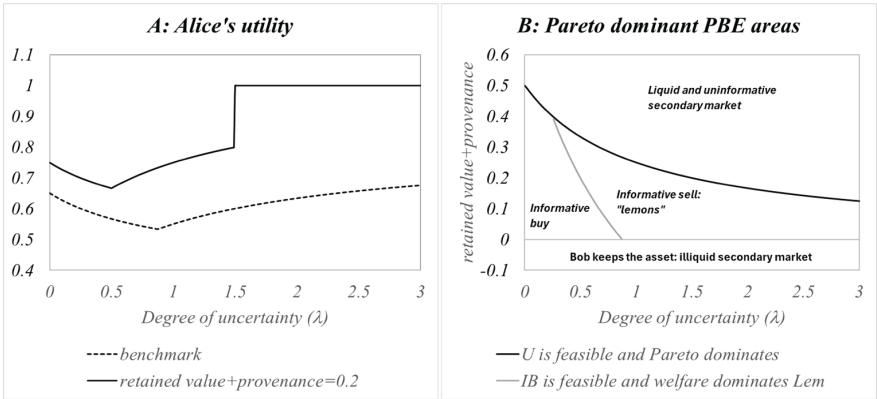

Figure 1: PBE: Tokenization and secondary market.

uncertain assets, Bob's initial purchase is informative: observing that Bob bought the asset reveals favorable information. For intermediate uncertainty, resale instead has a lemons-market interpretation: Bob's decision to sell is bad news. When the uncertainty is low, Bob always buys and resells regardless of his signal. In this case, the market is fully liquid but uninformative.

These effects are illustrated on Figure 1.[3] Since lower λ corresponds to greater uncertainty, moving from left to right in Panels A and B of Figure 1 corresponds to moving from highly uncertain assets to assets with more predictable common values. This distinction maps naturally into the asset classes we consider: venture capital and speculative art are closer to the low-λ region, while mature real estate is closer to the high-λ region.

Panel A shows how the highest achievable equilibrium utility varies with λ. When λ is low, Bob's initial purchase is a valuable and informative signal that increases the asset value. While the asset may not always be bought in the primary (the primary market is thin), Alice's expected gains combine the positive signaling effect of Bob's initial purchase with the value created by ownership records. Interestingly, some reduction in uncertainty in this region actually reduces Alice's payoff. This is because the informative-buy equilibrium still relies on selling only when Bob has a high signal, but the value of that high signal

[3]The figure assumes that $\bar{v} = 0.5$ and $\bar{z} = 0.3$. Panel A further assumes that retained value plus provenance is $\delta_B + \varsigma_B = 0.2$.

is lower when prior uncertainty is lower. As λ increases, Bob's posterior valuation after a high signal, $\frac{\lambda\bar{v}+1}{\lambda+1}$, moves closer to the prior mean $\bar{v}$, so Alice can extract less from the same separating pricing strategy that induces only high-signal Bob to buy.

As λ increases further, the nature of the primary and secondary market changes. Bob's initial purchase stops revealing information, and instead resale becomes the informative event. The same secondary market technology can therefore produce either positive signaling or a lemons market, depending on the uncertainty of the asset. For high-uncertainty assets, tokenization can make early ownership publicly informative: who bought first matters. For lower-uncertainty assets, tokenization can instead make exit publicly suspicious: why is the current owner selling? Thus the same public record that supports certification in venture-capital-like or speculative art settings can support adverse-selection concerns in more predictable assets such as ordinary real estate.

The fully liquid region is perhaps the most counterintuitive. When $\delta_B + \varsigma_B$ is sufficiently high, Bob buys and resells regardless of his signal. This is strictly the best region for Alice in payoff terms, but it is also the least informative secondary market regime. Once both types of Bob resell, resale no longer separates good news from bad news. Therefore, the model separates two notions: a successful secondary market need not be an informative one. Full liquidity can be valuable precisely because digital ownership records create enough resale value, but that same liquidity eliminates the informational content of trade.

Panel B summarizes these forces by showing how $\delta_B + \varsigma_B$ and λ jointly determine the type of secondary-market interaction. Small positive values of $\delta_B + \varsigma_B$ support partial liquidity, with positive signaling at low λ and lemons dynamics at higher λ. Larger values make resale attractive for both signal types, generating a fully liquid but uninformative market. Thus, the vertical axis determines whether digital records create enough resale value, while the horizontal axis determines what resale means informationally.

The framework maps naturally into different types of indivisible assets. For art and NFTs, the central question is whether provenance, visible ownership history, community benefits, or retained status make $\delta_B + \varsigma_B$ sufficiently positive. If not, exclusivity alone is

not enough to support valuable resale. For private equity and venture-capital-like assets, λ is likely to be low, so the main channel is endogenous information revelation through the decisions of informed early buyers; retained reputational benefits may further support trade. For real estate, λ is likely to be higher, and $\delta_B + \varsigma_B$ may be positive only in special cases, such as celebrity ownership, historically significant buildings, or architecturally distinctive properties. In ordinary real estate, prior use may instead make δ_B weak or negative, limiting the value of tokenized secondary markets.

Overall, Proposition 3.2 and Figure 1 suggest that tokenization produces different likely market dynamics for different tokenized assets rather than a universal liquidity improvement. Tokenization changes not only whether assets trade, but what trading means.

4 Token design

The baseline model shows that tokenization creates value only when secondary trade interacts with an economic mechanism: non-rival ownership benefits, provenance, information revelation, or incentives. This section studies how token design changes these mechanisms.

The first design separates usage rights from financial ownership rights, allowing provenance and usage value to be traded independently from claims on the common financial value. The second design fractionalizes ownership, allowing the ownership record itself to generate endogenous membership or network benefits. The third design introduces royalties, which affect secondary-market trade and, when Alice can take post-sale actions, can strengthen her incentives to create value after the initial sale. These design features are alternative ways in which tokenization changes the allocation of rights, the information content of trade, and the incentives of the original creator.

The same framework also applies to tokenized claims associated with sustainable firms, prestige goods, and luxury assets. In these settings, tokenization may create value not only through financial payoffs, but also through verifiable association with a firm, project, brand, or ownership community. For sustainable firms, digital ownership records may generate par-

ticipation, visibility, or identity benefits for investors who value being publicly associated with a mission-oriented project. For prestige and luxury assets, the relevant channels are provenance, exclusivity, status, and retained association with prior ownership. These examples highlight why the economic value of tokenization depends on whether digital records create non-rival benefits or alter incentives, rather than on tradability alone.

4.1 Separation of usage and financial rights

The first token-design choice is whether usage rights and financial ownership rights are bundled in a single token or traded separately.

In the baseline model, the same token gives its holder both the usage value $\bar{z}$ and the claim on the uncertain common value v. Suppose instead that Alice creates two separate tokens. The usage token is sold at price p_A^z and gives the holder the usage value. The ownership token is sold at price p_A^v and gives the holder the claim on the common value v.

We continue to refer to the first buyer as Bob, but the user of the asset and the financial owner need not be the same agent. This distinction is natural in real estate, where one agent may hold the financial claim while another rents or uses the property. It is also natural in art, where the holder of the usage right may keep, display, or showcase the artwork. In private equity, the analogue is less literal, but one can interpret the usage token as control, access, or other private-benefit rights, while the ownership token represents cash-flow rights.

Assume that provenance is associated with the usage token rather than the ownership token, as this is the more natural and economically interesting case that allows us to highlight the key distinctions relative to the one-token setting.[4] Thus, if Carol buys the usage token from Bob, her usage value is $z_C = \bar{z} + \delta_B$, while if she buys it directly from Alice, her usage value is $z_C = \bar{z}$. We allow δ_B to be positive or negative. A positive δ_B captures value-enhancing provenance, while a negative δ_B captures cases in which prior use or association reduces the value of later use. The ownership token carries the claim on v, Bob continues to

[4]Provenance could alternatively be attached to the ownership token. We attach it to the usage token because this gives the less mechanically positive and more informative separation case.

observe the signal s_B about v, and Bob may retain value $\varsigma_B \geq 0$ after selling the ownership token.

The following proposition summarizes the PBE implications. The ownership-token result follows directly from Proposition 3.2; the usage-token result follows from the separate pricing of the usage.

Proposition 4.1. *Suppose that Alice issues a usage token and an ownership token separately, with prices p_A^z and p_A^v. The following equilibria give Alice the highest payoff.*
(1) Alice sets the price of the usage token to $p_A^z = \bar{z} + \max\{0, \delta_B\}$. Bob resells the usage token if and only if $\delta_B \geq 0$, and sets the resale price $p_B^z = \bar{z} + \delta_B$. Alice's payoff from the usage token is therefore $u_A^z = \bar{z} + \max\{\delta_B, 0\}$.
(2) For the ownership token, the PBE are those in Proposition 3.2 after setting $\bar{z} = 0$ and $\delta_B = 0$. Alice's total payoff under separate tokens is $u_A^{sep} = u_A^z + u_A^v$, where u_A^v is determined by ς_B and λ.
(2a) If $0 \leq \varsigma_B \leq \frac{1-\bar{v}}{\lambda+1}$, there exists a threshold

$$\bar{\lambda}^v \equiv \frac{\bar{v} - (1-\bar{v})\varsigma_B}{(1-\bar{v})(\bar{v}+\varsigma_B)},$$

which determines whether the PBE that gives Alice the highest payoff involves an informative buy equilibrium or a lemons market equilibrium. Alice's total expected utility is

$$u_A = \begin{cases} u_A^{IB,sep} = \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \varsigma_B\right) + \bar{z} + \max\{\delta_B, 0\}, & \text{if } \lambda \leq \bar{\lambda}^v, \\ u_A^{Lem,sep} = \frac{\lambda\bar{v}}{\lambda+1} + \varsigma_B + \bar{z} + \max\{\delta_B, 0\}, & \text{if } \lambda \geq \bar{\lambda}^v. \end{cases}$$

(2b) If $\varsigma_B \geq \frac{1-\bar{v}}{\lambda+1}$, the market is liquid but uninformative. Alice's total expected payoff is

$$u_A = u_A^{U,sep} = \bar{v} + \varsigma_B + \bar{z} + \max\{\delta_B, 0\}.$$

Proof. Alice's optimal pricing of the usage token is immediate: Bob resells if and only if

$\bar{z} + \delta_B - p_A^z \geq \bar{z} - p_A^z$, i.e., if and only if $\delta_B \geq 0$, so Alice extracts $u_A^z = \bar{z} + \max\{\delta_B, 0\}$. The ownership-token result value u_A^v follows directly from Proposition 3.2. □

Proposition 4.1 shows three ways in which separate tokens affect the value of tokenization. First, separate tokens can shield the ownership token from negative provenance associated with use. In the one-token model, provenance and retained value enter jointly through $\delta_B + \varsigma_B$. Hence, sufficiently negative provenance can prevent secondary trade even when Bob retains some value from past ownership. With separate tokens, negative provenance affects the usage token, but it does not directly reduce the resale value of the ownership token.

Second, separate tokens can generate differential liquidity across the two components of the asset. The usage token is sold with certainty at price $\bar{z} + \max\{\delta_B, 0\}$, while the ownership token may still be sold only to a high-signal Bob, traded through a lemons market, or traded in an uninformative liquid market.

Third, separation can also reduce the feasibility of some resale equilibria for the ownership token. Once usage value and provenance are removed from the ownership token, the ownership-token resale condition depends only on ς_B rather than on $\delta_B + \varsigma_B$. Thus, when provenance is positive, separating the tokens makes the fully liquid uninformative ownership-token equilibrium harder to sustain.

Figure 2 illustrates these effects.[5] Panel A shows that, even without an active secondary market, separate tokens can increase Alice's payoff when uncertainty is sufficiently high. This is because the usage token can be sold separately even when the ownership token is not always bought by Bob. Thus, separation can create value through differential primary-market liquidity across the usage and ownership claims.

Panel B shows that adding the secondary-market channel further increases Alice's payoff when retained value and provenance make resale valuable. The comparison is useful because it separates two effects: the gain from issuing usage and ownership tokens separately, and the additional gain from allowing the ownership token to trade in a secondary market.

[5]The figure assumes that $\bar{v} = 0.5$ and $\bar{z} = 0.3$. Panel A further assumes that $\varsigma_B = 0$. The other panels maintain $\delta_B + \varsigma_B = 0.2$ in the corresponding one-token comparison.

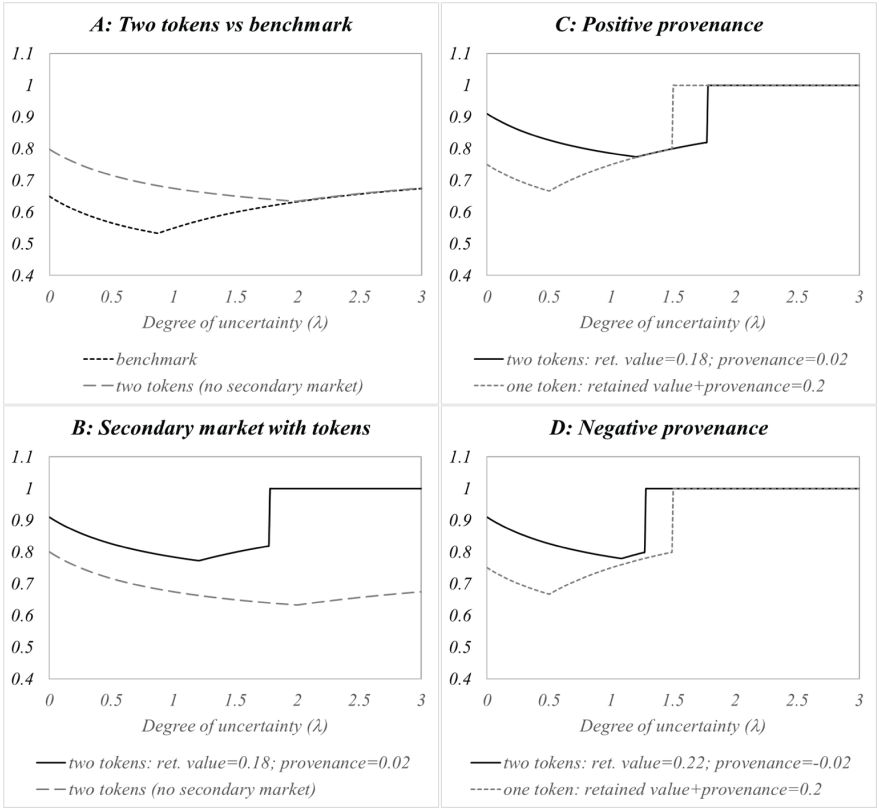

Figure 2: The value and effect of separating the token rights.

Panels C and D compare separate tokens with the one-token case from Section 3.3, shown by the dotted line. The comparison shows that separating usage and ownership changes the thresholds at which different information dynamics yield the highest payoff to Alice. In the one-token case, provenance and retained value enter jointly through $\delta_B + \varsigma_B$. With separate tokens, provenance affects the usage token, while only ς_B supports resale of the ownership token. As a result, positive provenance no longer helps to sustain the uninformative liquid equilibrium for the ownership token. This makes the fully liquid ownership-token equilibrium harder to reach, which can lower Alice's payoff relative to the one-token case.

Panel D illustrates the opposite side of the same tradeoff. When provenance attached to use is negative, separate tokens shield the ownership token from this negative effect. Negative provenance reduces the value of the usage token, but it does not directly distort

the informational resale of the ownership token. This case is particularly relevant for real estate, where prior use may reduce later usage value, while the financial claim on future cash flows or possible repurposing need not be affected in the same way.

The figure also highlights why uncertainty matters. At low values of λ, the informative-buy equilibrium is important, and separate tokens can be especially valuable because Alice can monetize the usage component even when the ownership component is sold only after a favorable signal. This mechanism is likely less important when usage or enjoyment benefits are small relative to financial value, as in many private-equity applications. It may be more relevant for tokenized art, where uncertainty about common value can be high while usage, display, enjoyment, and provenance effects remain economically meaningful. Thus, perhaps somewhat surprisingly, the separation of usage and ownership rights may be especially relevant for art-like assets, even though tokenized real estate is often the context in which this separation is most emphasized.

Overall, Figure 2 shows that separate tokens do not simply dominate or underperform bundled tokenization. They isolate the usage/provenance channel from the financial-information channel. This can raise value by creating differential liquidity and shielding the ownership token from negative provenance, but it can also reduce value when positive provenance would otherwise help sustain a liquid one-token secondary market. A model with uncertainty is needed to reveal these effects, because the value of separation depends not only on what is traded, but also on what buying and resale decisions reveal.

This logic also applies beyond the asset classes discussed above. For sustainable firms or social enterprises, separating rights may allow financial claims to be distinguished from participation or recognition benefits. For example, one token could represent a cash-flow or investment claim, while another could represent membership in a community of supporters, access to sustainability reports or events, or visible association with the firm's mission. In our model, these non-financial benefits enter through usage or enjoyment value rather than through the common financial value v.

Prestige and luxury tokens provide a related example. A token linked to a luxury brand,

collectible, or scarce experience may combine a financial claim with access, status, or authentication benefits. Separating these rights can be useful when the financial value and the prestige value are held by different agents. It can also matter when provenance is attached to use, display, or public association rather than to the financial claim itself. Thus, the separation of usage and financial rights is valuable when tokenization allows different components of value to be allocated to the agents who value them most.

4.2 Fractional ownership and endogenous ownership-records

This extension provides a complementary interpretation of the value of digital ownership records as a source of non-rivalry. Namely, instead of considering historical records that reflect the character of Bob, the first buyer, we consider the possibility that such non-rivalry may arise endogenously under fractional ownership and associated network effects.

To study this, assume that Alice creates a set of tokens consisting of a lead-investor claim, held by Bob, and K identical co-investor claims. To maintain comparability with the baseline model, each fractional claim bundles ownership and usage/enjoyment values. The lead-investor claim gives Bob a share $1-\gamma$ of the asset, while the co-investor claims together represent a share γ of the asset and are allocated equally to K identical uninformed co-investors. These co-investors each hold a fractional share $\frac{\gamma}{K}$ of the asset. As before, Alice sets p_A to monetize the asset, i.e., p_A is the price for the full set of tokens and equals the tokens' immediate market capitalization if both Bob and all co-investors buy. Consequently, Bob pays $(1-\gamma)\,p_A$ for his token and each co-investor pays $\frac{\gamma}{K}p_A$ for a co-investor token.

Bob observes $s_B \in \{0,1\}$ as in the baseline model. To keep this extension comparable with the main model, assume that the co-investors do not observe private signals. However, they observe Bob's purchasing decision, which may reveal information about v, the common financial value. The co-investors' beliefs are denoted by $\mu_K = \mathbb{E}\left[v|1_{AB}=1\right]$, where 1_{AB} denotes Bob's decision to buy.

We assume that fractional ownership can generate additional value through the ownership base. We capture this by an endogenous ownership-record term $\delta(K)$, instead of the

exogenous provenance term δ_B in the main model. This term reflects value created by verifiable participation in the ownership base, such as community benefits, visibility, or network effects. We assume that $\delta(0) = 0$ and that $\delta(K)$ is hump-shaped. At low levels of ownership dispersion, $\delta(K)$ is increasing in K, reflecting these benefits. At higher levels of dispersion, however, additional co-investors may reduce value because the ownership base becomes diluted and the tokens lose the exclusivity of a membership club.

For concreteness, we assume

$$\delta(K) = \eta \left(K - \frac{1}{2\psi} K^2 \right), \tag{9}$$

with $\eta > 0$ and $\psi > 0$, where ψ is a positive integer. This functional form captures network or community benefits of joint ownership, which are stronger when η is higher and diminish more slowly when ψ is higher. The co-investor claims are interpreted as membership claims that remain in place during the short-run secondary-market stage. Bob may resell the lead-investor claim to Carol, while we show that co-investors retain their claims in equilibrium. This preserves the ownership base that generates $\delta(K)$.[6]

To make the setting richer, we assume that Bob still obtains retained value $\varsigma_B \geq 0$ if he resells his lead-investor claim to Carol. This is because the lead investor is most likely to be associated with having "discovered" Alice and obtained information about her project. At the same time, the membership benefit $\delta(K)$ is generated by the continuing ownership base and is obtained by the co-investors and by Carol if they buy Bob's lead-investor claim. Since co-investors do not obtain Bob's retained benefit ς_B, Alice may not be able to incorporate this retained benefit fully into the initial price. This creates an endogenous cost of issuing co-investor claims.

Finally, we allow Alice to choose K optimally and treat γ as fixed.[7] The following proposition summarizes the optimal choices and feasible PBE that maximize Alice's payoff,

[6]This modelling choice keeps the size of the ownership base fixed, and hence the associated network effect, unchanged. An equivalent alternative would be to introduce a new buyer for each co-investor.

[7]The share parameter γ scales individual payments and claims, but does not affect Alice's total revenue from selling the full tokenized asset package.

conditional on Alice issuing the fractionalized token structure.

Proposition 4.2. *Provided that Alice issues a token to Bob and K co-investors, it is always optimal for her to set $K^* = \psi$, such that $\delta(K^*) = \frac{\eta\psi}{2} > 0$. The following summarizes the outcomes under PBE that give her the highest utility. In all these equilibria Bob resells his share to Carol with positive probability and co-investors choose to keep their share of tokens.*
(1) If $0 \leq \varsigma_B + \dfrac{\eta\psi}{2} \leq \dfrac{1-\bar{v}}{\lambda+1}$ and $\varsigma_B \leq \dfrac{\bar{v}}{\lambda+1}$, the best PBE for Alice involve partially liquid primary or secondary markets. There exists a threshold

$$\bar{\lambda}^{frac} \equiv \frac{\bar{v} - \left((1-\bar{v})\left(\bar{z} + \frac{\eta\psi}{2}\right) + \varsigma_B\right)}{(1-\bar{v})\left(\bar{z} + \frac{\eta\psi}{2} + \bar{v}\right) + \varsigma_B}, \tag{10}$$

which determines whether the equilibrium involves informative buying and resale, or a lemons market. Alice's total expected utility is

$$u_A = \begin{cases} u_A^{IB,frac} = \bar{v}\left(\dfrac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \dfrac{\eta\psi}{2}\right), & \text{if } \lambda \leq \bar{\lambda}^{frac}, \\ u_A^{Lem,frac(1)} = \dfrac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \dfrac{\eta\psi}{2} + \varsigma_B, & \text{if } \lambda \geq \bar{\lambda}^{frac}. \end{cases} \tag{11}$$

(2) If $0 \leq \varsigma_B + \dfrac{\eta\psi}{2} \leq \dfrac{1-\bar{v}}{\lambda+1}$ and $\varsigma_B \geq \dfrac{\bar{v}}{\lambda+1}$, the best PBE for Alice involves a lemons market and her utility is

$$u_A = u_A^{Lem,frac(2)} = \bar{v} + \bar{z} + \frac{\eta\psi}{2}. \tag{12}$$

(3) If $\varsigma_B + \dfrac{\eta\psi}{2} \geq \dfrac{1-\bar{v}}{\lambda+1}$, the best PBE for Alice is an uninformative equilibrium and her utility is

$$u_A = u_A^{U,frac} = \bar{v} + \bar{z} + \frac{\eta\psi}{2}. \tag{13}$$

Proof. See Appendix A.2. □

The key implication of Proposition 4.2 is that fractional ownership can generate non-rival benefits endogenously. Instead of relying on Bob-specific provenance, the value of the ownership record comes from the size and composition of the ownership base. This requires

positive ownership-base effects, captured by $\eta > 0$. Conditional on issuing fractionalized tokens, Alice chooses the number of co-investors to maximize this ownership-base effect. Under the assumed functional form, the optimal number is $K^* = \psi$, and the resulting ownership-base value is $\eta\psi/2$. Thus, Alice issues more co-investor claims when network effects diminish more slowly, corresponding to higher ψ.

The second key message of Proposition 4.2 is that the equilibrium outcomes maintain a similar structure in terms of equilibrium information revelation across feasible PBE and the role of asset-specific uncertainty in determining, which of these equilibria are the most desirable for Alice.

There are two subtleties. First, Alice may not be able to extract Bob's retained benefit fully through the initial price. The reason is that the same token package must also be attractive to uninformed co-investors, who do not receive Bob's retained value ς_B. This is why ς_B does not enter Alice's payoff in the informative-buy equilibrium or in the uninformative liquid equilibrium. This effect is small when retained benefits are small relative to the other sources of asset value, as may be the case for many tokenized private-equity or art applications. It can be more important for art and NFTs, where association with early ownership may generate substantial retained value.

Second, when ς_B is sufficiently large, the uninformative liquid equilibrium becomes easier to sustain and the adverse-selection cost of the lemons equilibrium is reduced. A larger retained benefit makes Bob more willing to buy even after a low signal. It also makes co-investors less exposed to the negative inference from Bob's resale decision, because Bob's participation is supported partly by retained value rather than only by the asset's common value.

Finally, as it is ultimately Alice's token design choice whether to issue a non-divisible token to informed Bob or to fractionalize it, it is useful to compare the results in Proposition 4.2 with those in Proposition 3.2 of the main setting. In the informative-buy equilibrium and in the uninformative liquid secondary market equilibrium, fractional ownership is better for Alice whenever $\eta\psi/2 \geq \delta_B + \varsigma_B$, that is, whenever the maximal ownership-base effect exceeds

the Bob-specific non-rival value in the one-token structure. The exogenous provenance δ_B should here be interpreted as the marginal value added of Bob's sole ownership record of the asset. In the lemons-market region, fractional ownership is more likely to dominate the one-token structure. If Bob's retained value is low, which occurs when $\varsigma_B \leq \dfrac{\bar{v}}{\lambda+1}$, fractional ownership is better for Alice whenever $\eta\psi/2 \geq \delta_B$. If Bob's retained value is high, i.e. $\varsigma_B \geq \dfrac{\bar{v}}{\lambda+1}$, the corresponding condition is $\eta\psi/2 \geq \delta_B + \varsigma_B - \bar{v}/(\lambda+1)$. These conditions show that fractional ownership can mitigate the adverse effects of lemons-market resale when ownership-base benefits are sufficiently strong. Both conditions are easier to satisfy, and thus, fractional ownership mitigates the negative lemons equilibrium effects whenever a better equilibrium is not there.

Sustainable-firm tokens provide a natural interpretation of this mechanism. A broad ownership base may create value when token holders derive utility from visible participation in a mission-oriented project, such as a green-energy firm, a conservation project, or a social-impact venture. In this case, the ownership record creates a community or recognition benefit that is not reducible to the common financial value of the firm. This corresponds to a positive ownership-base effect $\delta(K)$. The hump-shaped specification captures the idea that some ownership dispersion can increase visibility and community value, but excessive dispersion may dilute exclusivity, weaken identity, or reduce the perceived value of membership.

Prestige and luxury tokens provide a parallel interpretation. Fractional ownership of a luxury good, collectible, or branded asset may create value when the ownership base itself functions as a membership club. A small group of token holders may derive status or access benefits from verifiable participation. However, if too many co-investors are admitted, the token may lose exclusivity. This is precisely the tradeoff captured by the hump-shaped $\delta(K)$.

4.3 Royalties and Alice's post-sale effort

The final token-design feature considered is a royalty on secondary sales. Royalties differ from the previous design choices because they do not directly change which rights are traded. Instead, they change the division of surplus from secondary-market transactions and may

affect Alice's incentives after the initial sale. We continue to assume that Alice sells one bundled token and first consider the case in which she cannot affect the asset's value after the initial sale.

Proposition 4.3. *Suppose that Alice can set a royalty $r \in [0, 1)$, but cannot take a value-increasing action after the initial sale. For any fixed $r > 0$, relative to the no-royalty case, Alice's utility is either the same or strictly lower. The latter occurs in the case of the "lemons" equilibrium, where Bob always buys the asset, but resells it only when he received a low signal, $s_B = 0$. Furthermore, the conditions for PBE with higher payoffs for Alice are harder to satisfy whenever $r > 0$; hence, setting $r = 0$ is weakly optimal for Alice.*

Proof. See Appendix A.1. □

Proposition 4.3 shows that basic royalties either do not help or strictly reduce Alice's utility. This is because a royalty is effectively a transaction fee on the secondary trade, and, as we have shown, equilibria with secondary trading yield higher utility to Alice when feasible. We have also argued in Section 2 that Alice can substitute the royalty with a higher initial price that anticipates Bob's gains from trade in the secondary market, making it payoff-neutral in most feasible equilibria. The reason why the "lemons" equilibrium yields strictly lower utility is that Alice collects the royalty only with probability $1 - \bar{v}$, i.e., when Bob observes a low signal $s_B = 0$, while the royalty reduces Bob's overall expected payoff. The technical details are in Appendix A.1.

Let us now consider an extension of the model where the possibility of setting a royalty can be value-increasing. Namely, consider that, despite having sold the tokenized asset, Alice can take a costly action to increase the asset value. For example, in the case of art, Alice could increase the token holder's enjoyment value from ownership by creating a related special event, exhibition, or membership club, and inviting the token holder to participate. In the case of venture capital, Alice could continue to exert effort in developing her invention despite having given up the ownership rights, thereby increasing the asset's long-term value.

Formally, we keep the one-token environment and the timing of Section 3.3, but add one

post-sale action. After Bob has decided to resell the asset, and before he sets the resale price, Alice can choose effort $e_A \geq 0$. This effort increases Carol's valuation of the asset by $\Delta = \alpha e_A$, where $\alpha > 0$, and costs Alice $\varkappa(e_A)$, where $\varkappa'(e_A) > 0$ and $\varkappa''(e_A) > 0$. For the subsequent analysis, we assume $\varkappa(e_A) = \bar{\varkappa} e_A^2/2$, with $\bar{\varkappa} > 0$. Alice receives the royalty share r of the resale price and therefore chooses effort only when resale occurs. Bob and Carol anticipate this effort when deciding whether to resell and buy, respectively.

Proposition 4.4. *Suppose that Alice can increase the asset value before Carol buys it. Then, for any royalty $r \in [0, 1)$, Alice chooses $e_A^* = r\alpha/\bar{\varkappa}$, which induces an increase in the asset value by $\Delta^* = \kappa r$, where $\kappa \equiv \alpha^2/\bar{\varkappa}$.*
Conditional on the informative-buy equilibrium being feasible, Alice's payoff is

$$u_A^{IB}(r) = \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \varsigma_B + \kappa r - \frac{\kappa r^2}{2}\right),$$

and is strictly increasing in r. Conditional on the uninformative-resale equilibrium being feasible, Alice's payoff is

$$u_A^{U}(r) = \bar{v} + \bar{z} + \delta_B + \varsigma_B + \kappa r - \frac{\kappa r^2}{2},$$

and is strictly increasing in r. Thus, in both equilibria, Alice chooses the highest royalty consistent with the relevant PBE constraints.
Conditional on the lemons equilibrium being feasible, Alice's utility is

$$u_A^{Lem}(r) = (1 - r\bar{v})\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \kappa r\right) + \varsigma_B - (1-\bar{v})\frac{\kappa r^2}{2}.$$

Hence, the optimal royalty in the lemons equilibrium is zero whenever Alice's effort technology is not sufficiently productive to overturn the secondary-trade distortion, i.e., $\kappa \leq \bar{v}\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B\right)$.
Alice does not make an effort if there is no royalty or if Bob does not buy and resell.

Proof. See Appendix A.3. □

Proposition 4.4 shows that royalties can matter once Alice can affect the asset value after sale. In the informative-buy and uninformative-resale equilibria, a higher royalty increases Alice's effort incentives and raises her utility, subject to the PBE constraints that keep resale feasible. In the lemons equilibrium, the transaction cost dominates the value increase as long as Alice's post-sale effort is not the main driver of the asset value (see Appendix A.3 for details).

The key logic of Proposition 4.4 is straightforward: royalties can add value when they encourage Alice to make effort ex post. Alice is more easily incentivized to do so when the relevant resale equilibrium involves either an informative-buy equilibrium or an uninformative equilibrium with a liquid market.

The optimal royalty therefore depends on the equilibrium region and on the feasibility constraints, which are affected by r. The main implication is that the optimal royalty need not be monotone in asset-specific uncertainty. A positive royalty may be optimal in low-uncertainty or high-uncertainty regions, where the relevant equilibrium is informative buying or fully liquid resale, but may be zero in the lemons region when post-sale effort is not sufficiently productive. At the same time, a high royalty can make otherwise valuable resale equilibria infeasible by weakening Bob's resale incentives.

Taken together, these examples show that token design matters most when tokenization changes the non-financial attributes of ownership. Sustainable-firm tokens may be valuable when they create verifiable participation, community, or mission-alignment benefits. Prestige and luxury tokens may be valuable when they create status, authentication, provenance, or membership benefits. In both cases, the relevant economic question is not whether the asset can be traded on a blockchain, but whether the tokenized ownership record creates non-rival value, improves allocation across heterogeneous users, reveals information, or strengthens incentives.

5 Empirical predictions

The model generates asset-specific empirical predictions. These predictions follow do not follow from tokenization per se, but from the interaction among four economic forces: public ownership records, non-rival benefits from past ownership or membership, uncertainty about common value, and usage or enjoyment value. The same tokenization technology can therefore generate different market outcomes across asset classes. It may produce informative primary-market purchases, lemons-type resale, fully liquid but uninformative secondary markets, or no economically meaningful secondary market at all. The relevant empirical question is not whether an asset has been tokenized, but which economic channel tokenization activates.

Art tokens. Art tokens are a natural setting in which digital ownership records have first-order effects on value. The model predicts that secondary-market activity should be strongest when provenance, membership-club value, and retained ownership benefits are large. In contrast, tokens whose value relies only on digital scarcity or exclusivity should exhibit weak resale incentives, as pure exclusivity does not encourage active trading. Furthermore, as some art assets are likely to involve noticeable uncertainty, we expect this market to often exhibit limited trade in the primary market, as in the informative-buy equilibrium.

As non-fungible tokens (NFTs) are a natural example of art tokens, these observations are broadly consistent with the stylized fact that most NFTs are never traded, while a small subset achieves very high valuations. Prominent examples include Bored Ape Yacht Club and CryptoPunks, as well as individual works such as Beeple's *Everydays: The First 5000 Days*, which sold at Christie's for $69.3 million in 2021. These cases map naturally into the model: early buyers are visible, ownership is publicly verifiable, and ownership may confer status, community access, or historical provenance. Bored Ape Yacht Club is especially close to the membership-club interpretation, as ownership has historically been associated with access to communities, events, airdrops, and related benefits. CryptoPunks instead illustrates the provenance channel more directly: much of its value appears linked to historical significance, early adoption, and the visibility of ownership of one of the earliest NFT collections.

These patterns are also consistent with the empirical work mentioned in the introduction. For example, Oh et al. (2022) documents strong following behavior in NFT markets, where investors are more likely to buy assets that have recently attracted attention or have been purchased by others. While this is often interpreted as herding, our model suggests an alternative and complementary interpretation: observed buying can be rationally informative when early buyers are perceived to have superior information, status, or curatorial value. Our model can also rationalize OpenSea's decision to stop enforcing royalties in 2023. Indeed, royalties reduce liquidity in the secondary market and are only valuable when this is offset by maintained values later on, for example through new perks, access, exhibitions, or community benefits. Absent this effect, as in the case of typical NFTs, the artist would typically be better off charging a higher primary-market price and not hampering secondary-market liquidity.

Prestige and luxury tokens. Prestige and luxury tokens provide a closely related setting in which ownership records may be valuable even when the underlying asset is not primarily financial. Examples include tokens linked to luxury goods, scarce collectibles, exclusive experiences, or branded membership communities. In the language of the model, these assets are likely to have substantial usage or enjoyment value, and their value may depend strongly on provenance, authentication, and visible association with prior owners or with a particular brand.

The model also predicts a tradeoff between exclusivity and liquidity. Prestige value often depends on scarcity, but secondary-market liquidity requires that ownership transfer creates surplus. Thus, luxury-token markets should be most active when exclusivity is combined with transferable or persistent status benefits, rather than when exclusivity is purely rival. Fractionalized luxury tokens may be valuable when the ownership base functions as a membership club, but excessive dispersion can dilute status. This maps directly into the hump-shaped ownership-base effect $\delta(K)$.

Private-equity tokens. Private-equity and venture-capital tokens are also characterized by a notable degree of uncertainty about common value. For these assets, the model predicts that informed early buyers play a central role in price formation. Participation by

sophisticated investors could act as a positive signal, while the absence of such participation should significantly reduce demand. This market could also exhibit lemons dynamics. The model therefore provides structure for empirical work testing whether measures of uncertainty are associated with informative-buy or lemons-market dynamics.

Royalties and other programmable contractual features could be relevant in this setting if they help preserve founder, sponsor, or early-investor incentives after partial exit. In the model, royalties are valuable only when they induce post-sale value creation. Applied to private-equity or venture-capital-like tokens, this suggests that tokenization may relax the liquidity-control tradeoff when programmable claims credibly preserve incentives for key agents to remain involved. The relevant prediction is therefore conditional: tokenized private claims should be more valuable when secondary liquidity is paired with contractual features that sustain effort, governance, or monitoring incentives.

For some private-equity tokens, the structure with a lead investor and uninformed co-investors is also realistic and commonly observed in equity crowdfunding. While many current equity-crowdfunding structures do not allow secondary trading, should tokenization enable it, the model predicts that lead-investor resale need not necessarily be bad news, especially when uncertainty is high, and retained benefits are sufficiently large. Furthermore, under these conditions lead investors are more likely to sell than co-investors.

Real-estate tokens. Real estate is likely to involve lower uncertainty, as cash flows and asset values are typically more predictable. The model therefore predicts that early purchases are less informative, while resale is more likely to raise adverse-selection concerns when the asset is not fully standardized. A sharper empirical prediction is that lemons-type outcomes are more likely when the asset still involves a noticeable degree of uncertainty, while less uncertain real-estate assets could be liquid in both primary and secondary markets.

That said, provenance and retained value are harder to justify in this context, which can make this asset class less likely to benefit from tokenization through ownership records alone. Positive provenance is likely to matter only for specific assets, such as landmark properties, historically significant buildings, or celebrity-owned real estate. However, when

some provenance effects are present, this is also the market where separating usage and ownership tokens could be most valuable, as separation can shield ownership tokens from deterioration of usage value. Furthermore, if there are positive network effects from co-owning a famous property, fractionalized ownership can also be value-adding.[8]

More broadly, the model suggests that tokenization should not be expected to generate deep secondary markets for standard real-estate assets unless ownership records create additional value or other mechanisms support liquidity.

Sustainable firm tokens. Beyond the aforementioned assets, sustainable firms and social enterprises provide an additional case for tokenization. In sustainable firms and social enterprises, token holders may value not only financial claims, but also the nonpecuniary utility from supporting a firm that generates social or environmental benefits, such as access to healthcare or education, clean-energy projects, climate-oriented ventures, or other explicitly impact-oriented enterprises. This implies that as per our theory tokenization may create value through persistent and observable ownership records that can make this support visible beyond the holding period. This creates retained ownership value when early investors continue to receive reputational, affiliation, or status benefits after resale. It can also create provenance value when later buyers value tokens previously held by prominent impact investors, foundations, public figures, or respected institutions. Ceteris paribus, we expect that tokens linked to high-visibility missions should command higher prices, attract broader participation, and exhibit stronger resale activity than otherwise similar claims with opaque ownership histories. Resale premia should be especially likely when prior owners are credible or mission-aligned, and when the firm is young, difficult to evaluate, or operates where social impact is hard to verify.

The informativeness and liquidity of these markets should vary with uncertainty and token design. For risky early-stage social enterprises or innovative sustainable firms, uncertainty about financial returns and impact may be high, so purchases by informed or reputable investors may be good news, producing markets that are thin but informative.

[8]Outside the mechanisms considered in our model, fractionalization may also be valuable in real estate because many investors cannot afford to buy an entire property; see Dai et al. (2025).

For mature solar-energy assets, established healthcare-access firms, or education platforms with observable performance, resale may be more liquid but less informative, because trades may reflect affiliation, portfolio rebalancing, or liquidity needs rather than new information. Fractionalization should create value when it generates membership benefits, visibility, public engagement, or collective participation, but not when ownership becomes so diffuse that affiliation and signaling value are diluted. Royalties should be valuable only when the originator's post-sale effort affects future value, for example through impact verification, community engagement, or credibility maintenance; otherwise they merely tax resale. Thus, sustainable-firm and social-enterprise tokens should be most active and valuable when ownership records affect affiliation, certification, impact visibility, community value, or post-sale incentives, and may remain inactive when these mechanisms are absent.

6 Conclusion

This paper develops a theory of when tokenization creates value for indivisible real-world assets. The central message of our model is that tokenization does not add value merely because it makes an asset technically easier to trade. Nor are common token features, such as royalties or fractionalization, necessarily valuable. Tokenization creates value only when its technological features change the economics of ownership: by generating non-rival ownership benefits, making ownership histories verifiable, improving the allocation of usage and financial rights, revealing information through observed trading decisions, or strengthening post-sale incentives.

The model studies an originator, an informed first buyer, and later secondary-market buyers. This structure captures a broad set of tokenized assets, including art, collectibles, private-equity and venture-capital-like claims, real estate, prestige and luxury assets, and sustainable-firm and social-enterprise tokens. The first buyer may have private information about the asset's common financial value, while later buyers observe the first buyer's actions through the ownership record. Digital records can therefore affect both valuations and beliefs.

They may create retained value for the first buyer after resale, provenance value for later buyers, or membership and network benefits under fractional ownership.

The analysis delivers two negative results. First, pure tradability is not sufficient. If past ownership is value-neutral or value-reducing, resale may not occur even when the token is technically transferable. Second, royalties are not automatically beneficial. When the originator cannot affect asset value after the initial sale, royalties are either payoff-neutral or distort secondary-market trade. Royalties become valuable only when they improve the originator's incentives to create post-sale value.

When digital ownership records do create non-rival benefits, tokenization can support several distinct market regimes. For highly uncertain assets, early purchases by informed buyers can be favorable public signals. For intermediate levels of uncertainty, resale can instead generate lemons-market dynamics, because the decision to sell may reveal bad information. When provenance or retained ownership benefits are sufficiently strong, secondary markets can become fully liquid but uninformative: trade occurs regardless of the first buyer's signal. Thus, the model separates liquidity from information. A successful secondary market need not be an informative one.

Token design matters because different designs activate different economic channels. Separating usage and financial rights can improve allocation and shield financial ownership from negative provenance attached to use. Fractional ownership can create endogenous ownership-record benefits when a broad ownership base generates community, visibility, or club value, although excessive dispersion may dilute exclusivity. Royalties can support value creation when they preserve the creator's, founder's, or sponsor's post-sale incentives. These features are therefore not independent add-ons; they are alternative ways in which tokenization changes rights, records, beliefs, and incentives.

The model also generates empirical predictions. Tokenized assets should trade more actively when past ownership is valuable, verifiable, and salient, as in art, collectibles, prestige goods, sustainable firms, social enterprises, celebrity-linked assets, or ventures backed by prominent investors. Early purchases should be especially informative for assets with sub-

stantial uncertainty, such as speculative art or venture-capital-like claims. Resales should be discounted when they are interpreted as exits following unfavorable information. For more predictable assets, such as ordinary real estate or social enterprises in healthcare or education, tokenization should create less incremental liquidity unless ownership records generate unusual provenance value, usage and ownership rights can be separated, or fractional ownership creates membership benefits. Sustainable-firm or social-enterprise tokens should be more valuable when digital records create verifiable participation, identity, or community benefits rather than merely representing transferable financial claims.

More broadly, the paper provides a framework for studying value creation whenever ownership histories become observable, portable, and priced. A positive provenance effect means that prior ownership can raise future value rather than merely transfer existing claims. A negative provenance effect captures the opposite force: prior use, undesirable association, or reputational contamination can reduce value. Retained ownership benefits and provenance effects can create a value chain across sequential buyers, in which each owner's identity and actions become part of the asset's economic history. The value of tokenization therefore depends not on the existence of a blockchain record alone, but on whether that record changes how the asset is valued, traded, and interpreted.

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A Proofs

A.1 Proof of Proposition 3.2 and 4.3

. To economize on space, we present the proof for the more general setting with royalties and allow for the possibility that Carol's usage value includes an additional constant term $\Delta \geq 0$ beyond the provenance effect should she buy from Bob. This formulation is useful for the later extensions that endogenize Δ. Throughout the proof, $r \in [0,1]$ and Δ are treated as fixed constants, and Proposition 3.2 is obtained as the special case with $r = 0$ and $\Delta = 0$.

Bob's pure strategies after observing his signal $s_B \in \{0,1\}$ are denoted as follows. Let $1_{AB}(s_B) \in \{0,1\}$ denote Bob's decision to buy the asset from Alice, $1^K_{AB}(s_B) \in \{0,1\}$ his decision to keep the asset conditional on buying it, and $1^S_{AB}(s_B) \in \{0,1\}$ his decision to offer it for resale to Carol. We solve the game by backward induction and verify Perfect Bayesian Equilibrium. A PBE specifies strategies and Carol's beliefs after observable histories (we denote beliefs with $\tilde{1}$ when highlighing beliefs in important). Strategies are sequentially rational given these beliefs, and beliefs are Bayesian on path. Off path, we use pessimistic beliefs: deviations by Bob are not interpreted as positive certification of the asset. To keep notation compact, we spell out beliefs only where needed.

Since Carol can buy from Bob only after Bob has bought the asset and offered it for resale, her on-path belief following that history is $\mu_{BC} = \mathbb{E}\left[v|\tilde{1}_{AB}(s_B) = 1, \tilde{1}^S_{AB}(s_B) = 1\right]$. Similarly, Carol can buy directly from Alice only if Bob does not buy, in which case her on-path belief is $\mu_{AC} = \mathbb{E}\left[v|\tilde{1}_{AB}(s_B) = 0\right]$.

From Carol's utility (4), allowing for the additional constant Δ, Carol buys from Bob at price p_B if, and only if, $\mu_{BC} + \bar{z} + \delta_B + \Delta - p_B \geq 0$. Because Bob sets the resale price, he extracts all of Carol's surplus and thus chooses

$$p^*_B = \mu_{BC} + \bar{z} + \delta_B + \Delta. \tag{14}$$

If Bob buys and keeps the asset, his utility (2) is $u_B(s_B)|_{1_{AB}(s_B)=1, 1^K_{AB}(s_B)=1} = \frac{\lambda\bar{v}+s_B}{\lambda+1} + \bar{z} - p_A$.

If instead he buys and resells, his utility (3) is $u_B(s_B)|_{1_{AB}(s_B)=1,\, 1^S_{AB}(s_B)=1} = (1-r)p^*_B + \varsigma_B - p_A$. Hence Bob weakly prefers resale to keeping the asset if and only if

$$(1-r)(\mu_{BC} + \delta_B + \Delta) + \varsigma_B - r\bar{z} \geq \frac{\lambda\bar{v} + s_B}{\lambda + 1}. \tag{15}$$

In each candidate equilibrium, Alice sets the primary-market price to extract Bob's surplus conditional on $1_{AB}(s_B) = 1$. Thus, whenever Bob buys because resale is valuable, the relevant participation constraint is determined by $u_B(s_B)|_{1_{AB}(s_B)=1, 1^S_{AB}(s_B)=1}$ rather than by $u_B(s_B)|_{1_{AB}(s_B)=1, 1^K_{AB}(s_B)=1}$, while he buys to keep if resale is not optimal. The conditions for buying to (always) keep and Alice's pricing strategy and utility are equivalent to those in Lemma 3.1.

Because $u_B(s_B)|_{1_{AB}(s_B)=1, 1^K_{AB}(s_B)=1}$ is strictly increasing in s_B, monotonicity implies that any pure-strategy PBE *with positive-probability resale* must take one of these three forms:

(i) $1_{AB}(1) = 1$, $1_{AB}(0) = 0$, and $1^S_{AB}(1) = 1$, i.e., Bob only buys and resells if $s_B = 1$. We call it the "informative buy" (IB) equilibrium.

(ii) $1_{AB}(s_B) = 1$ for any s_B, $1^S_{AB}(1) = 0$, $1^S_{AB}(0) = 1$, i.e., Bob always buys, but only resells if $s_B = 0$. We call it the "lemons" (Lem) equilibrium.

(iii) $1_{AB}(s_B) = 1$ and $1^S_{AB}(s_B) = 1$ for any s_B, i.e., Bob always buys and resell. We call it the "uninformative" (U) equilibrium.

Case (i). In this candidate PBE the history in which Carol meets Bob occurs if and only if $s_B = 1$, while the history in which Carol can buy from Alice occurs if and only if $s_B = 0$. Bayes' rule therefore implies $\mu_{BC} = \mathbb{E}[v|s_B = 1] = \frac{\lambda\bar{v}+1}{\lambda+1}$ and $\mu_{AC} = \mathbb{E}[v|s_B = 0] = \frac{\lambda\bar{v}}{\lambda+1}$.

Sequential rationality of Bob's resale decision for type $s_B = 1$ requires

$$(1-r)(\delta_B + \Delta) + \varsigma_B \geq r\left(\frac{\lambda\bar{v} + 1}{\lambda + 1} + \bar{z}\right). \tag{16}$$

Conditional on resale, Bob sets $p^*_B = \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \Delta$. Since Bob only buys if $s_B = 1$, it is optimal for Alice to extract Bob's surplus by setting $p^*_A = (1-r)p^*_B + \varsigma_B$, or equivalently $p^*_A = (1-r)\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \Delta\right) + \varsigma_B$. Alice's expected utility in this candidate equilibrium

is $u_A^{IB} = \bar{v}(p_A^* + rp_B^*)$, which simplifies to

$$u_A^{IB} = \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \Delta + \varsigma_B\right).$$

Moreover, if Bob does not buy, Carol's belief on-path is $\mu_{AC} = \frac{\lambda\bar{v}}{\lambda+1}$, and she does not buy as long as $\mu_{AC} + \bar{z} - p_A^* < 0$, i.e., whenever

$$r < \frac{\frac{1}{\lambda+1} + \delta_B + \Delta + \varsigma_B}{\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \Delta}.$$

Hence, the feasibility of this equilibrium requires that the royalty is not prohibitively high. It is easy to verify that when $r = 0$, $\mu_{AC} + \bar{z} - p_A^* \leq 0$ for any $\delta_B + \Delta + \varsigma_B \geq 0$. It is also worth noting that u_A^{IB} does not depend on the royalty r, and even if Alice is able to set a positive royalty, zero royalty remains weakly optimal under these beliefs. Finally, a low-signal Bob who deviates by buying and reselling induces the same on-path belief $\mu_{BC} = \mathbb{E}[v|s_B = 1]$. At $p_A^* = (1-r)p_B^* + \varsigma_B$, this deviation gives him zero payoff, equal to his outside option. Hence the proposed strategy is sequentially rational under the tie-breaking convention that an indifferent low-signal Bob stays out.

Case (ii). Under these strategies and beliefs Bob's resale reveals that $s_B = 0$, and by Bayes' $\mu_{BC} = \mathbb{E}[v|s_B = 0] = \frac{\lambda\bar{v}}{\lambda+1}$. Sequential rationality further requires that Bob must prefer to sell the asset if $s_B = 0$, and by (15) it must hold that

$$(1-r)(\delta_B + \Delta) + \varsigma_B \geq r\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z}\right).$$

At the same time, it must also hold that Bob with signal $s_B = 1$ prefers to keep the asset, i.e., by (15)

$$(1-r)(\delta_B + \Delta) + \varsigma_B \leq \frac{1}{\lambda+1} + r\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z}\right).$$

It follows that this equilibrium can only exist with positive probability if λ is finite, i.e., when there is some uncertainty about the asset value. Conditional on resale, Bob sets

$p_B^* = \frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \Delta$. Should Bob resell a "lemon" his utility is $u_B^{Lem} = (1-r)p_B^* + \varsigma_B - p_A$. As Alice sets the price, on-path consistency and payoff optimization requires that

$$p_A^* = \min\left[(1-r)p_B^* + \varsigma_B, \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}\right].$$

When the resale participation constraint binds, this gives

$$p_A^* = \frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \Delta + \varsigma_B - r\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \Delta\right),$$

which holds as long as

$$(1-r)p_B^* + \varsigma_B \leq \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}.$$

Equivalently,

$$\delta_B + \Delta + \varsigma_B - r\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \Delta\right) \leq \frac{1}{\lambda+1}.$$

Furthermore, assuming that the above condition holds, Alice's utility is $u_A^{Lem} = p_A^* + (1-\bar{v})\, rp_B^*$

$$u_A^{Lem} = (1-r\bar{v})\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \Delta\right) + \varsigma_B.$$

Case (iii). Finally, all Bob's decisions may be uninformative on path and Bayes' rule implies $\mu_{BC} = \mathbb{E}[v] = \bar{v}$. Sequential rationality requires both types of Bob to prefer resale to keeping the asset. Since the condition (15) is tighter for $s_B = 1$, it is necessary and sufficient that

$$(1-r)(\delta_B + \Delta) + \varsigma_B \geq \frac{1-\bar{v}}{\lambda+1} + r(\bar{v} + \bar{z}).$$

Conditional on resale, Bob sets $p_B^* = \bar{v} + \bar{z} + \delta_B + \Delta$. Since Bob always buys and resells in equilibrium, Alice sets

$$p_A^* = (1-r)p_B^* + \varsigma_B = (1-r)(\bar{v} + \bar{z} + \delta_B + \Delta) + \varsigma_B.$$

Alice's expected utility is therefore

$$u_A^U = \bar{v} + \bar{z} + \delta_B + \Delta + \varsigma_B,$$

which is again independent of r.

Finally, it is clear from the above that when $(1-r)(\delta_B+\Delta)+\varsigma_B < 0$ no positive-probability resale PBE of the forms considered above can be sustained; the outcome therefore reduces to a no-resale/keep equilibrium, with Alice's pricing problem as in the benchmark.

Welfare comparisons. As highlighted in Section 2 the main welfare criterion of interest is Alice's utility to create the asset. We are mainly interested in equilibria that give Alice the highest payoff, and thus compare Alice's utility under $r = 0$, for the welfare implications of Proposition 3.2. The benchmark equilibrium yield the lowest utility for Alice and the uninformative equilibrium yields the highest one. Furthermore, the informative-buy (IB) equilibrium yields a higher utility than the lemons equilibrium if and only if $\bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \Delta + \varsigma_B\right) \geq \frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \Delta + \varsigma_B$, which is equivalent to

$$\lambda \leq \frac{\bar{v} - (1-\bar{v})(\bar{z} + \delta_B + \Delta + \varsigma_B)}{(1-\bar{v})(\bar{z} + \bar{v} + \delta_B + \Delta + \varsigma_B)}. \tag{17}$$

Thus, as in the benchmark, the informative-buy equilibrium is more attractive to Alice when uncertainty is sufficiently high. Next, $u_A^U - u_A^{Lem} = \bar{v}+\bar{z}+\delta_B+\Delta+\varsigma_B - \left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \Delta + \varsigma_B\right)$. Hence, conditional on feasibility, the fully liquid equilibrium weakly dominates the lemons equilibrium. Finally, the fully liquid equilibrium dominates the informative-buy equilibrium whenever $\bar{v} + \bar{z} + \delta_B + \Delta + \varsigma_B \geq \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \Delta + \varsigma_B\right)$. This condition is more likely to hold when provenance, retained value, and Δ are sufficiently large, so that inducing resale by both types is particularly valuable. Taken together, these inequalities characterize the parameter regions in which each candidate pure-strategy PBE is feasible and, among feasible equilibria, preferred by Alice.

Hence, whenever the corresponding feasibility and incentive conditions hold, the stated strategy profile together with the Bayes-consistent on-path beliefs and pessimistic off-path

beliefs constitutes a PBE.

Finally, setting $r = 0$ and $\Delta = 0$ yields Proposition 3.2. In that case, (15) reduces to $\mu_{BC} + \delta_B + \varsigma_B \geq \frac{\lambda\bar{v}+s_B}{\lambda+1}$. Accordingly, the informative-buy equilibrium with resale requires $\delta_B + \varsigma_B \geq 0$, the lemons equilibrium requires $0 \leq \delta_B + \varsigma_B \leq \frac{1}{\lambda+1}$, and the fully liquid uninformative resale equilibrium requires $\delta_B + \varsigma_B \geq \frac{1-\bar{v}}{\lambda+1}$. The lemons equilibrium may continue to exist when $\delta_B + \varsigma_B \in \left[\frac{1-\bar{v}}{\lambda+1}, \frac{1}{\lambda+1}\right]$, but in that region the fully liquid uninformative equilibrium is feasible and yields a weakly higher payoff to Alice. These are the corresponding conditions underlying Proposition 3.2.

A.2 Proof of Proposition 4.2

Let us first consider that the number of token holders, K, is fixed and that Alice aims to sell the asset to Bob and K co-investors.

The plausible PBE and Carol's beliefs that support these equilibria remain as in Appendix A.1. The key drivers of Bob's resale decision also remain the same as in Appendix A.1. Namely, using the assumption in Section 4.2, Carol is willing to buy Bob's asset share at

$$p_B^* = \mu_{BC} + \bar{z} + \delta(K),$$

and Bob is willing to resell his asset share if

$$\mu_{BC} + \delta(K) + \varsigma_B \geq \frac{\lambda\bar{v} + s_B}{\lambda + 1}.$$

We also need to consider how co-investors' beliefs affect their participation and resale decisions.

Let us start with the informative-buy equilibrium, where Bob participates if and only if $s_B = 1$. Because Bob's buying decision is informative for both Carol and the co-investors, it must hold that $\mu_K = \mu_{BC} = \frac{\lambda\bar{v}+1}{\lambda+1}$. Each individual co-investor's value of keeping the asset is therefore $\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta(K)$, and co-investors cannot gain from selling it. In fact, if

their off-path beliefs are such that the overall community benefit is lower whenever even one co-investor deviates to selling, then they strictly prefer not to sell the asset.[9]

At the same time, Bob is willing to buy the asset from Alice if $(1-\gamma)p_A \leq (1-\gamma)\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta(K) + \varsigma_B\right)$, and each co-investor is willing to buy from Alice if $\frac{\gamma}{K}p_A \leq \frac{\gamma}{K}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta(K)\right)$. Consequently, if Alice aims to sell all tokens, the co-investors' participation constraint binds and she sets $p_A = \frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta(K)$. Her expected utility is therefore

$$u_A^{IB,frac}(K) = \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta(K)\right).$$

Similarly to before, feasibility of this PBE requires $\delta(K) + \varsigma_B \geq 0$.

Let us then consider the lemons and uninformative-liquid equilibria, where Bob participates for both signal realizations. Bob's participation is then uninformative at the primary-market stage, and the co-investors' belief is $\mu_K = \mathbb{E}[v] = \bar{v}$. In the lemons equilibrium, Carol and the co-investors learn s_B only after Bob has decided whether to keep or resell his asset share. Namely, they do not know s_B when they decide whether to participate in the primary market and by the law of iterated expectations, $\mu_K = \mathbb{E}\left[\mathbb{E}[v|s_B]\right] = \bar{v}$.

In the lemons equilibrium, Bob resells only after the low signal and sets $p_B^* = \frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta(K)$. Bob is willing to buy the asset from Alice if $(1-\gamma)p_A \leq (1-\gamma)\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta(K) + \varsigma_B\right)$, while each co-investor is willing to buy if $\frac{\gamma}{K}p_A \leq \frac{\gamma}{K}\left(\bar{v} + \bar{z} + \delta(K)\right)$. Therefore, if Alice wants to sell all tokens, she sets $p_A = \bar{z} + \delta(K) + \min\left\{\frac{\lambda\bar{v}}{\lambda+1} + \varsigma_B, \bar{v}\right\}$, and her utility is

$$u_A^{Lem,frac}(K) = \bar{z} + \delta(K) + \min\left\{\frac{\lambda\bar{v}}{\lambda+1} + \varsigma_B, \bar{v}\right\}.$$

Similarly to before, this equilibrium requires $\delta(K) + \varsigma_B \geq 0$, so that Bob is willing to resell after $s_B = 0$, and $\delta(K) + \varsigma_B \leq \frac{1}{\lambda+1}$, so that Bob prefers to keep the asset after $s_B = 1$.

In the uninformative-liquid equilibrium, Bob buys and resells for both signal realizations, and Bob's participation decision is uninformative. Bob sets $p_B^* = \bar{v} + \bar{z} + \delta(K)$. He is willing

[9] While this could be formalized more explicitly, the argument is straightforward and we do not present it to save space.

to buy the asset from Alice as long as $(1-\gamma)p_A \leq (1-\gamma)\left(\bar{v}+\bar{z}+\delta(K)+\varsigma_B\right)$, while the co-investors are willing to buy as long as $\frac{\gamma}{K}p_A \leq \frac{\gamma}{K}\left(\bar{v}+\bar{z}+\delta(K)\right)$. The co-investors' participation constraint therefore binds, and Alice's utility is

$$u_A^{U,frac}(K) = \bar{v}+\bar{z}+\delta(K).$$

For this PBE to exist, Bob must be willing to resell the asset even after $s_B = 1$, which gives $\delta(K)+\varsigma_B \geq \frac{1-\bar{v}}{\lambda+1}$.

Next, consider Alice's choice of K. Since her utility in all candidate equilibria above is increasing in $\delta(K)$, it follows that $K^* = \arg\max_K \delta(K)$, where $\delta(K)$ is defined in (9 and ψ is assumed to be an integer, the community effect is maximized at

$$K^* = \psi,\ \delta(K^*) = \frac{\eta\psi}{2}.$$

Hence Alice always chooses $K^* = \psi$, and the relevant comparison is across equilibrium branches evaluated at this value.

The uninformative-liquid equilibrium is feasible at K^* if and only if $\varsigma_B + \frac{\eta\psi}{2} \geq \frac{1-\bar{v}}{\lambda+1}$. Whenever this condition holds, the highest utility Alice can obtain is $u_A^{U,frac} = \bar{v}+\bar{z}+\frac{\eta\psi}{2}$. Moreover, whenever the uninformative-liquid equilibrium is feasible, this payoff weakly dominates the informative-buy and lemons payoffs.

Suppose instead that

$$\varsigma_B + \frac{\eta\psi}{2} < \frac{1-\bar{v}}{\lambda+1}.$$

Then the uninformative-liquid equilibrium is infeasible for all K. Since $\frac{1-\bar{v}}{\lambda+1} < \frac{1}{\lambda+1}$, the lemons equilibrium remains feasible at K^*. The informative-buy equilibrium is also feasible at K^* because $\varsigma_B + \frac{\eta\psi}{2} < \frac{1-\bar{v}}{\lambda+1}$ implies the relevant resale condition is below the threshold at which Bob would resell after the high signal. Alice therefore compares

$$u_A^{IB,frac} = \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1}+\bar{z}+\frac{\eta\psi}{2}\right)$$

and

$$u_A^{Lem,frac} = \bar{z} + \frac{\eta\psi}{2} + \min\left\{\frac{\lambda\bar{v}}{\lambda+1} + \varsigma_B, \bar{v}\right\}.$$

There are two cases. First, if $\varsigma_B \geq \frac{\bar{v}}{\lambda+1}$, then the co-investors' participation constraint binds in the lemons equilibrium and $u_A^{Lem,frac} = \bar{z} + \frac{\eta\psi}{2} + \bar{v}$. This payoff strictly dominates the informative-buy payoff, since $\bar{z} + \frac{\eta\psi}{2} + \bar{v} > \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \frac{\eta\psi}{2}\right)$. In this region, the payoff-maximizing feasible branch is the lemons equilibrium. Second, if $\varsigma_B < \frac{\bar{v}}{\lambda+1}$, then Bob's low-signal participation constraint binds in the lemons equilibrium and $u_A^{Lem,frac} = \bar{z} + \frac{\eta\psi}{2} + \frac{\lambda\bar{v}}{\lambda+1} + \varsigma_B$. In this case, the lemons equilibrium gives Alice a higher payoff than the informative-buy equilibrium if and only if $\bar{z} + \frac{\eta\psi}{2} + \frac{\lambda\bar{v}}{\lambda+1} + \varsigma_B \geq \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \frac{\eta\psi}{2}\right)$. Equivalently, the lemons equilibrium dominates the informative-buy equilibrium if and only if

$$\lambda \geq \bar{\lambda}^{frac} \equiv \frac{\bar{v} - \left((1-\bar{v})\left(\bar{z} + \frac{\eta\psi}{2}\right) + \varsigma_B\right)}{(1-\bar{v})\left(\bar{z} + \frac{\eta\psi}{2} + \bar{v}\right) + \varsigma_B}.$$

If this inequality is reversed, the informative-buy equilibrium gives Alice the higher payoff.

To summarize, Alice sets $K^* = \psi$, using this in utility expressions in this appendix gives (11), (12, and (13), which proves the proposition.

A.3 Proof of Proposition 4.4

This proof builds on Appendix A.1 and endogenizes Δ, which was taken as fixed there. Namely, let $\Delta = \alpha e_A$, where e_A is Alice's costly effort to increase the asset value after Bob has decided to resell. Provided that Alice has set a royalty r and Bob resells the asset, Alice chooses effort to solve

$$e_A^* = \arg\max_{e_A}\left\{rp_B^* - \frac{\bar{\varkappa}e_A^2}{2}\right\} = \arg\max_{e_A}\left\{r\alpha e_A - \frac{\bar{\varkappa}e_A^2}{2}\right\},$$

where p_B^* is given by (14). It follows that $e_A^* = \frac{r\alpha}{\bar{\varkappa}}$. Let us define $\kappa \equiv \frac{\alpha^2}{\bar{\varkappa}}$, which implies that $\Delta^* = \alpha e_A^* = \kappa r$. Under Alice's optimal effort, Bob's resale price is therefore

$$p_B^* = \mu_{BC} + \bar{z} + \delta_B + \kappa r.$$

The effort cost Alice incurs conditional on resale is

$$C_A \equiv \frac{\bar{\varkappa}(e_A^*)^2}{2} = \frac{\kappa r^2}{2}.$$

Thus, relative to Appendix A.1, the only changes are that Δ is replaced by κr and Alice's utility subtracts the expected effort cost in the states in which Bob resells.

First, consider the informative-buy equilibrium, where Bob buys only if $s_B = 1$ and resells after buying. Then $\mu_{BC} = \frac{\lambda\bar{v}+1}{\lambda+1}$, and initial purchase and resale occurs with probability $\bar{v}$. Alice's utility is

$$u_A^{IB}(r) = \bar{v}\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z} + \delta_B + \varsigma_B + \kappa r - \frac{\kappa r^2}{2}\right).$$

As $\frac{\partial u_A^{IB}(r)}{\partial r} = \bar{v}\kappa(1-r) > 0$ for all $r \in [0,1)$, conditional on this equilibrium being feasible, Alice chooses the highest feasible royalty.

The feasibility conditions are obtained from Appendix A.1 by considering that $\Delta = \kappa r$. Bob's willingness to make the asset available for resale requires that

$$(1-r)(\delta_B + \kappa r) + \varsigma_B \geq r\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}\right).$$

Moreover, as Bob's does not buy on path when $s_B = 0$, Carol's belief after Bob does not buy is $\mu_{AC} = \frac{\lambda\bar{v}}{\lambda+1}$. Sequential rationality also requires that Carol does not buy directly from Alice, which gives

$$(1-r)\left(\delta_B + \kappa r\right) + \varsigma_B \geq r\left(\frac{\lambda\bar{v}+1}{\lambda+1} + \bar{z}\right) - \left(\frac{1}{\lambda+1}\right).$$

The first condition is clearly tighter.

Second, consider the lemons equilibrium, where Bob always buys and resells only if $s_B = 0$. Then $\mu_{BC} = \frac{\lambda\bar{v}}{\lambda+1}$. Following Appendix A.1, Alice's utility is

$$u_A^{Lem}(r) = (1 - r\bar{v})\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B + \kappa r\right) + \varsigma_B - (1-\bar{v})\frac{\kappa r^2}{2}.$$

Differentiating,

$$\frac{\partial u_A^{Lem}(r)}{\partial r} = -\bar{v}\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B\right) + \kappa - \kappa r(1+\bar{v}).$$

Hence, if $\kappa \leq \bar{v}\left(\frac{\lambda\bar{v}}{\lambda+1} + \bar{z} + \delta_B\right)$, then $\frac{\partial u_A^{Lem}(r)}{\partial r} \leq 0$ for all $r \in [0,1)$. Therefore, conditional on the lemons equilibrium, Alice sets $r^*_{Lem} = 0$.

Third, consider the uninformative resale equilibrium, where Bob always buys and always resells. Then $\mu_{BC} = \bar{v}$, and resale occurs with probability one. Alice's utility is

$$u_A^U(r) = \bar{v} + \bar{z} + \delta_B + \varsigma_B + \kappa r - \frac{\kappa r^2}{2}.$$

Thus $\frac{\partial u_A^U(r)}{\partial r} = \kappa(1-r) > 0$ for all $r \in [0,1)$. Hence, conditional on this equilibrium being feasible, Alice chooses the highest feasible royalty.

Using (15) and replacing Δ with κr, the relevant feasibility condition is

$$(1-r)(\delta_B + \kappa r) + \varsigma_B \geq \frac{1-\bar{v}}{\lambda+1} + r(\bar{v} + \bar{z}).$$

Thus the optimal royalty in the uninformative resale equilibrium is the supremum of the set of royalties satisfying this inequality. Combining the three cases establishes the result.